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A trustworthy cybersecurity model for transparent cyberattack detection using Bald Eagle Search tuned XGBoost and explainable AI

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Abstract

The rapid escalation of sophisticated cyberattacks demands intrusion detection systems that not only achieve high accuracy but also provide transparency and fairness in decision-making, addressing critical gaps in current machine learning-based cybersecurity solutions. This study proposes a trustworthy intrusion detection framework that integrates Bald Eagle Search (BES) optimization with the XGBoost classifier and SHAP-based explainability to enhance both predictive performance and interpretability. The BES algorithm efficiently fine-tunes key XGBoost hyperparameters to overcome limitations of traditional models that struggle with complex, high-dimensional intrusion patterns. Evaluated on a real-world cybersecurity dataset, the BES-optimized XGBoost model achieves a substantially improved accuracy of 99.7%, outperforming baseline XGBoost (87.5%) and other meta-heuristic optimizers including GA, ACO, and PSO. SHAP visualizations further reveal feature-level contributions, ensuring transparent and accountable detection decisions, while fairness analysis highlights disparities across browser types, promoting responsible and bias-aware cybersecurity deployment. Overall, the proposed model offers a robust, interpretable, and ethically aligned solution for detecting modern cyber threats, contributing significantly to the advancement of transparent and high-performance intrusion detection systems.

Keywords Cybersecurity, Intrusion detection, XGBoost, Bald Eagle Search optimization, SHAP explainability

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Introduction

The danger of cyber intrusions is greater than it has ever been before in a world where digital infrastructure supports every facet of our lives, including healthcare, banking, government, and communication. In order to evade conventional security measures, cybercriminals are getting craftier and craftier, using stealth tactics and zero-day vulnerabilities. Identifying and thwarting such incursions has grown in importance as organizations embrace remote work and highly linked systems. Firewalls, antivirus software, and intrusion prevention systems have come a long way, but many dangers still slip through the cracks. Because of this, we need to start using security measures that are smarter, more flexible, and aware of user behavior [1].

Using humanized AI models that detect threats by analyzing user behavior and network traffic flows over time, in addition to examining signatures or known patterns, is one of the most promising directions in cybersecurity defense. People have predictable habits when it comes to their digital behavior. For example, they often log in from the same places, open the same files, and interact with the same apps in the same ways. Unusual network activity, such as large data transfers, repeated access attempts, or unusual protocol usage, along with any deviation from these patterns, can signal a cyber intrusion [2–4].

Methods based on rules or signatures are the backbone of traditional intrusion detection systems (IDS). Attacks that are new or difficult to detect, like insider attacks or APTs, are beyond the capabilities of these methods because they rely on well-known entry points. Behavioral analytics and traffic analysis powered by AI, on the other hand, can adapt to new attack methods as they learn from data [5]. Artificial intelligence systems can mimic human intuition in their context-aware decision-making by integrating network traffic features (such as port usage, packet sizes, and connection durations) with behavioral cues (such as anomalies in login time or command execution patterns) [6, 7].

Consider the following scenario: a user logs in from an unfamiliar device at an odd hour, then proceeds to download files quickly and tries to steal data. An AI model trained on behavioral baselines would find this sequence extremely suspicious, even though individual actions might not trigger alarms. In addition to real-time anomaly detection, these systems can also prioritize threats according to risk scores, which helps security teams concentrate on the most important threats and reduces alert fatigue. Humanized AI systems also prioritize being open and easy to understand and explain. Analysts can gain faith in automated systems and facilitate prompt intervention by using techniques like SHAP (Shapley Additive exPlanations) to comprehend the rationale behind a specific behavior's flagging. By incorporating AI practices

that prioritize fairness, these systems can guarantee that decisions are made without favoring any one user or department over another [8–11].

In conclusion, a revolutionary step in cybersecurity has been made possible by combining analytics of network traffic with models of user behavior in order to detect cyber intrusions. Moving from reactive defenses to proactive, intelligent surveillance is a necessary evolution of security frameworks due to the increasing complexity and frequency of threats. Protecting our digital future will require humanized AI's technical prowess, ethical sensitivity, and adaptive reasoning [12–15].

Problem formulation

As cyber threats grow increasingly covert and adaptive, traditional security systems face limitations in detecting attacks that do not conform to predefined signatures. Human analysts often rely on intuition, context, and learned behavioural baselines to recognise unusual patterns. Emulating this capability, the challenge of detecting cyber intrusions based on network traffic and user behaviour can be formally framed as an anomaly detection and binary classification problem, grounded in human-aware, context-sensitive machine learning [16].

Let $X = \{x_1, x_2, \dots, x_n\}$ represent a dataset of observed activities, where each instance $x_i \in \mathbb{R}^d$ combines both network traffic features (e.g., packet count, port usage, connection duration) and user behaviour features (e.g., login times, access patterns, resource usage). Each instance may be labelled $y_i \in \{0, 1\}$, where 0 indicates normal behaviour and 1 suggests an intrusion.

The objective is to learn a function $f: \mathbb{R}^d \rightarrow \{0, 1\}$, such that:

$$\begin{aligned} f(x_i) &= 1, \text{ if } x_i \text{ is anomalous or indicative of intrusion} \\ f(x_i) &= 0, \text{ otherwise} \end{aligned}$$

This function should reflect trust, transparency, and context-awareness in humanised AI terms, not merely statistical deviation. Therefore, rather than relying purely on static thresholds, a probabilistic anomaly score $A(x_i) \in [0, 1]$ can be learned such that

$$A(x_i) = 1 - P(x_i | \mathcal{N}) \quad (1)$$

where $P(x_i | \mathcal{N})$ is the likelihood of instance x_i under the distribution of normal behaviour $\mathcal{N}$. An intrusion is flagged when $A(x_i) > \tau$, where τ is a learned or adaptive threshold.

To model user-specific behaviour, a user u_j 's historical baseline $B_{\{u_j\}}$ can be defined, and behaviour is anomalous if

$$\|x_i - B_{u_j}\| > \delta \quad (2)$$

where δ is the tolerance level based on learned variance across time. This multi-perspective formulation integrates network and behavioural domains, allowing humanised AI systems to not only detect threats but also offer explanations, adapt to evolving contexts, and prioritise fairness and interpretability in intrusion response.

The rise of sophisticated cyberattacks in modern digital infrastructures—spanning cloud platforms, IoT deployments, industrial systems, and enterprise services—has made intrusion detection a critical component of cybersecurity. Traditional rule-based and signature-driven intrusion detection systems (IDS) are increasingly ineffective as attackers adopt stealthier and more adaptive strategies, including zero-day exploits, polymorphic malware, and behavior-mimicking intrusions [17]. As a result, data-driven IDS models have gained attention, yet many suffer from key limitations such as sensitivity to hyperparameters, high false-negative rates, lack of interpretability, and inconsistent performance across diverse user or device environments. These shortcomings hinder reliable deployment in real-world, high-stakes cybersecurity applications.

Recent studies have explored deep learning [18], ensemble classifiers [19], and optimization-assisted IDS approaches [20]. While these models demonstrate performance gains, they typically address **only one dimension of trustworthiness**—accuracy, scalability, or robustness—while neglecting others. For example, hybrid deep learning models improve detection rates but remain opaque, limiting their utility in environments requiring explainability and auditability. Optimization-based models enhance classification accuracy but still lack interpretability or fairness evaluation. Likewise, fairness-aware or interpretable IDS remain rare, and existing works do not address the simultaneous need for transparency, security reliability, and bias-free decision-making. These gaps highlight the need for a multidimensional intrusion detection approach that integrates optimization, performance stability, explainability, and fairness in a unified framework.

To address these limitations, XGBoost stands out as an effective baseline for structured cybersecurity data, offering scalability and strong performance in classifying complex attack patterns [21]. However, its performance is highly dependent on hyperparameter tuning, and sub-optimal configurations frequently lead to overfitting or unstable behavior. This challenge has motivated the use of meta-heuristic optimization methods, yet commonly used algorithms such as GA, ACO, and PSO often suffer from premature convergence or require intensive parameter adjustments [21]. At the same time, the absence of explainable reasoning in most models restricts their usability in critical sectors—such as finance, industrial

control, and healthcare—where transparent justification of security decisions is essential.

Novelty and motivation

This study introduces a novel, trust-centered intrusion detection framework that integrates Bald Eagle Search (BES) optimization, XGBoost, and SHAP explainability in a unified pipeline to overcome limitations seen in existing IDS research. BES is employed not to generate interpretability itself, but to produce a more stable, better-generalized, and higher-performing XGBoost model, thereby enabling clearer and more meaningful SHAP-based explanations. SHAP, in turn, provides transparent feature-level reasoning for every prediction, addressing the black-box nature of prior models. Additionally, we incorporate fairness auditing across browser-type contexts, a critical yet underexplored factor in IDS deployments. This integration represents the first IDS framework that jointly optimizes detection accuracy, transparency, fairness, and model reliability—contributing a unique multidimensional perspective to trustworthy AI-driven cybersecurity.

Machine learning has demonstrated strong potential for enhancing intrusion detection, yet many models still suffer from limitations such as overfitting, poor generalization, and a lack of explainability—factors that restrict their deployment in mission-critical applications. Tree-based ensemble models like XGBoost have gained prominence in IDS research due to their robustness, scalability, and superior performance on structured network traffic data [22, 23]. XGBoost's gradient-boosted architecture enables it to capture non-linear relationships and subtle variations in traffic behavior, making it suitable for detecting both common and emerging cyberattacks. However, its performance is highly sensitive to hyperparameter selection, and suboptimal tuning can lead to degraded accuracy or bias in decision outcomes. To address this challenge, meta-heuristic optimization techniques have been increasingly adopted in cybersecurity to efficiently navigate high-dimensional search spaces and overcome the limitations of manual or grid-based tuning [24, 25].

Among these algorithms, the Bald Eagle Search (BES) optimizer offers several advantages, including a balanced exploration–exploitation mechanism, fewer control parameters, and efficiency in escaping local optima—making it well-suited for optimizing complex models like XGBoost. Compared to methods such as GA, PSO, or ACO, BES exhibits stronger stability and faster convergence when dealing with continuous optimization problems commonly encountered in machine learning hyperparameter tuning [26, 27]. Integrating BES with XGBoost therefore provides a compelling approach to

achieving a more accurate, reliable, and computationally efficient IDS framework.

While accuracy remains important, modern cybersecurity systems must also prioritize transparency to ensure that detection decisions can be interpreted, validated, and trusted by human analysts. Black-box models can hinder adoption in regulated or safety-critical domains. To overcome this challenge, SHAP (Shapley Additive Explanations) is incorporated to provide both global and instance-level interpretability, enabling stakeholders to understand how features such as failed logins, IP reputation scores, or browser behavior influence intrusion predictions [28, 29]. This interpretability is essential for enhancing user trust, facilitating forensic analysis, and meeting compliance requirements where explainable AI is mandatory.

In addition, fairness considerations have become increasingly important as intrusion detection models may inadvertently exhibit bias toward specific user groups or system attributes. Evaluating and improving fairness allows IDS systems to operate responsibly and equitably across diverse environments. Given these motivations, this study proposes a trustworthy and transparent cyber intrusion detection model that integrates BES-tuned XGBoost with SHAP explainability to deliver high detection accuracy, model interpretability, and fairness-aware intrusion analysis. The proposed framework addresses critical gaps in reliability, explainability, and adaptability that remain underexplored in current IDS research. There are the following research contributions of this work as below:

1. This study presents the first integration of Bald Eagle Search (BES) optimization with XGBoost for intrusion detection, addressing the shortcomings of poorly tuned boosting models in cybersecurity.
2. The proposed model incorporates a SHAP-based explainability layer, providing transparent, instance-level explanations and overcoming the black-box limitations of most IDS models.
3. The work includes a fairness-aware analysis, evaluating detection consistency across browser-type contexts—an aspect largely missing in existing intrusion detection research.
4. A multi-objective model design optimizes accuracy, interpretability, transparency, and fairness, significantly improving real-world deployment suitability.
5. Comparative experiments demonstrate that BES-tuned XGBoost outperforms GA, ACO, and PSO variants and achieves state-of-the-art accuracy while preserving interpretability.

While several IDS models have been developed using deep learning and meta-heuristic optimization, none have combined Bald Eagle Search (BES) optimization with XGBoost and SHAP-based explainability to simultaneously achieve high accuracy, interpretability, and fairness in cybersecurity detection. Existing literature primarily focuses either on performance enhancement or anomaly detection, but rarely integrates optimization + explainability + fairness analysis into a unified framework. This work introduces a first-of-its-kind BES-tuned XGBoost model capable of transparent cyberattack detection, supported by SHAP's fine-grained interpretation. In addition, the model performs fairness auditing across browser-type contexts, addressing a critical yet underexplored issue in IDS research. This multi-objective approach ensures a trustworthy, robust, and ethically reliable cybersecurity model that exceeds the capabilities of traditional black-box IDS systems.

The remaining paper is organized into following main sections. Section 2 reviews existing intrusion detection models and the role of nature-inspired algorithms in optimization. Section 3 explains the integration of Bald Eagle Search (BES) with XGBoost and SHAP for enhanced detection and interpretability. Section 4 presents comparative performance results, SHAP-based feature insights, fairness evaluation, and ablation analysis. Section 5 summarizes key outcomes and proposes directions for deploying the model in real-time and fairness-aware cybersecurity systems.

Related work

In order to detect cyberattacks in Industrial Control Systems, Al-Abassi et al. [30] create an ensemble deep learning framework. Their approach to ICS dataset class imbalance is a hybrid model that uses deep neural networks and decision tree classifiers. The method enhances detection across all classes by capturing the nuanced behavior of rare attack patterns. Compared to DNN, RF, and AdaBoost, it performed better on two ICS datasets evaluated with 10-fold cross-validation. When compared to other methods, the ensemble's accuracy and recall were the best.

Using Deep Denoising Belief Networks (DDBN) with both non-local and local regularization techniques, Guangming Xian [31] presents a semi-supervised deep learning approach. When labelled data is scarce, the model is designed for intrusion detection. According to the experimental results, DDBN outperformed GAN, SVM, and DBN-RFS in terms of training and testing errors. It allows for learning with little supervision to achieve high accuracy. Being unlabeled cyber threats, it is perfect for new ones.

An NIDS built on deep learning and anomaly detection with recurrent neural networks is presented by Md.

Mohiuddin Ahmed et al. (2025) [32]. The UNSW-NB15 and UGR datasets were used to assess the system. With an accuracy of up to 92.85% and a high degree of precision in detecting suspicious actions, the RNN model surpassed baseline models. Fast inference and low false positive rates were highlighted by the authors as important contributions. It demonstrates potential for detecting anomalies in high-speed networks in real-time.

For Internet of Things (IoT) networks, Oli and Arockiam (2017) [33] created a compact anomaly detection system utilizing MobileNet. Minimizing model complexity and resource consumption without sacrificing accuracy was their main goal. Their approach obtained a 98.4% accuracy rate while utilizing a small amount of memory on the NSL-KDD dataset. Internet of Things devices with limited resources will benefit greatly from this method. This makes it easier to deploy on platforms that use edge computing.

In their groundbreaking work, Zhang et al. (2022) [27] present an intrusion detection framework that utilizes Federated Learning. This framework safeguards user data privacy while effectively identifying cyber threats in distributed environments. Without transferring any raw data, the model makes use of numerous local learners integrated into a single model. Results demonstrated a privacy-preserving accuracy of 96.8% on the NSL-KDD dataset. In Internet of Things (IoT) settings spanning multiple domains, the system enables safe group study. It lessens the burden on NIDS applications to centralize data.

In order to detect network intrusions, K. L. Xie and P. Wang (2020) [26] suggest a CNN-LSTM model hybrid that can capture both the spatial and temporal aspects of traffic patterns. The model's application to the CICIDS2017 dataset resulted in a low false alarm rate and a detection accuracy of over 99%. Sequential learning and convolutional feature extraction both contribute to the hybrid architecture's success. With their framework, real-world traffic detection becomes faster and more accurate. This method is useful for complex enterprise networks because it allows for scalable deployment.

Stephen Jacob et al. [34] develop a microservices-based intrusion detection model using Graph Convolutional Networks. They apply a Diffusion Convolutional Recurrent Neural Network (DCRNN) to model API traffic traces for anomaly detection. The system identifies brute force, bot registration, and DDoS attacks by learning spatial-temporal trace dependencies. The proposed approach achieved high accuracy and low latency in detecting cyber intrusions within distributed systems. Their graph-based modeling approach outperformed traditional IDSs in microservices environments.

Kandhro et al. [25] develop a real-time cyber intrusion detection system for Industrial IoT using a GAN-based

architecture with Autoencoders and LSTM. The model detects malicious behaviors across various attack categories with impressive accuracy. Tested on NSL-KDD, KDDCup99, and UNSW-NB15, the model achieved 95–97% accuracy and high True Negative Rates. It offers a highly reliable and fast detection mechanism suited for smart industry applications. The system is notably efficient in edge-based IIoT environments.

Amit K. Shukla et al. [35] propose UInDeSI4.0, an unsupervised IDS tailored for Industry 4.0 using Isolation Forests and Principal Component Analysis. Evaluated on UNSW-NB15, the model reduced feature count to nine and achieved ~ 63% accuracy. It supports intrusion detection in unlabeled data environments with low computational overhead. Compared to supervised or deep learning methods, it offers faster performance while maintaining competitive detection rates in smart factories.

Shahid Allah Bakhsh et al. [36] introduce a deep learning-based IDS using FFNN, LSTM, and RandNN models to secure IoT networks. Tested on CIC-IoT22, FFNN reached 99.93%, LSTM 99.85%, and RandNN 96.42% accuracy. The approach combines pattern recognition and adaptive learning to offer rapid and high-accuracy detection for diverse IoT scenarios. It effectively handles complex traffic and evolving threats in large-scale IoT deployments.

S. Pavithra and K. V. Vikas [37] address the issue of unbalanced data in intrusion detection by combining XGBoost, LSTM, Mini-VGGNet, and AlexNet with Random Forests. Their hybrid model leverages ensemble learning and deep feature extraction to boost detection in minority attack classes. The approach also includes class balancing and optimal feature selection. Their results show significant improvement in detecting rare intrusions. The methodology proves effective in realistic, imbalanced cyber environments, where most attacks are underrepresented.

Asad Ali et al. [24] present GWOMLP-IDS, a Gray Wolf Optimization-enhanced MLP model for intrusion detection. GWO optimizes the MLP's weights and biases to improve generalization and minimize overfitting. Tested on KDD and UNSW datasets, the model achieved a detection accuracy of 99.37% and reduced the error detection rate compared to standard MLP models. Their approach improved intrusion detection precision while ensuring computational feasibility.

Rudri Kalaria et al. [38] introduce IoTPredictor, a Hidden Markov Model-based framework for predicting IoT device behavior and detecting malicious activities. By modeling device states—normal, compromised, counterfeit—the system enables early intrusion detection in fog environments. Experiments on simulated IoT traffic demonstrate robust prediction accuracy and effective

anomaly classification. The framework enhances the precision of IDS in resource-constrained IoT ecosystems.

Farhan Ullah et al. [39] propose IDS-INT, a transformer-based transfer learning IDS for imbalanced network traffic. They use feature extraction with CNN-LSTM and apply SMOTE for class balancing. Tested on UNSW-NB15, CIC-IDS2017, and NSL-KDD datasets, the model demonstrated improved detection of minority attacks. The explainable AI integration makes it suitable for trustable deployment in real-world NIDS environments.

Muhammad A. Othman Ahmed et al. [32] enhance IoT security using ensemble ML models, XGBoost and LightGBM. Their method analyzes RT-IoT2022 dataset features and applies SMOTE to handle class imbalance. XGBoost and LGBM reached 99.553% and 99.651% accuracy respectively. These classifiers showed strong generalization and minimized false positives, making them ideal for robust real-time IoT network security frameworks.

Shajjad Hossain et al. [40] design a privacy-preserving SSL-based IDS for 5G-V2X vehicular networks. Their model uses self-supervised pretraining and federated learning to reduce reliance on labeled data while preserving privacy. It achieved up to 9% accuracy improvement even with as few as 200 samples. The system is scalable and adaptive, enhancing intrusion detection for connected vehicles under constrained and private data settings.

Ying-Dar Lin et al. [41] propose DICI, a dual-model intrusion detection framework that integrates an IDS model with a Cyber Threat Intelligence (CTI) transfer model. DICI can dynamically incorporate new threat knowledge from CTI sources and apply it to real-time network data. The CTI transfer model enhances detection of novel threats not seen during training. On benchmark datasets, the framework improved the IDS F1-score by 9.29% and CTI-based detection by 30.92%. The model demonstrates real-time learning and adaptation to evolving cyber threats.

Modern intrusion detection research has evolved across three major directions—optimization-driven IDS, explainable IDS, and fairness-aware IDS. This section summarizes each theme to highlight existing progress and the specific gaps that motivate our proposed BES-tuned XGBoost framework with SHAP explainability and fairness auditing.

Optimization-based intrusion detection systems

A significant body of IDS research focuses on improving performance through meta-heuristic optimization techniques. Several works apply evolutionary algorithms such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Gray Wolf Optimization (GWO) to enhance feature selection

or classifier tuning. Ali et al. [24] used GWO to optimize MLP weights, achieving improved detection performance on KDD and UNSW datasets. Similarly, Kandhro et al. [25] proposed a GAN-Autoencoder-LSTM hybrid for IIoT environments, demonstrating robust performance across multiple datasets. GA-based feature selection techniques, also show improved accuracy but still suffer from convergence instability. While these approaches achieve gains in accuracy, they often lack reliable exploration–exploitation balance, making them susceptible to premature convergence—especially in high-dimensional hyperparameter spaces. Moreover, optimization-focused IDS models generally treat performance as the primary objective, with little emphasis on transparency or fairness. This gap motivates the use of **Bald Eagle Search (BES)** in our work, which offers smoother phase transitions, better global search stability, and stronger tuning performance compared to GA, PSO, and ACO, particularly for continuous hyperparameter optimization in XGBoost.

Explainable intrusion detection systems

Explainability has recently emerged as a core requirement for deploying IDS in real-world, high-stakes domains where black-box models are not acceptable. SPAFIS [42] uses soft prototypes to provide rule-based interpretability, while Sun and Wang [43] convert network traffic into images to interpret learned features through CNNs. Transformer-based IDS frameworks [39] integrate class balancing with attention-driven insights, offering partial interpretability. However, many high-performing models—such as deep learning-based or ensemble-based IDS (e.g., Ahmed et al. [32], Bakhsh [36])—lack transparent explanation mechanisms. Although SHAP and LIME have been explored in limited contexts, there remains a disconnect between **optimized models and interpretable reasoning**. No prior work combines **meta-heuristic hyperparameter optimization with SHAP-based explainability**, leaving a gap where models either perform well but remain opaque, or provide explanations but lack state-of-the-art accuracy. Our framework addresses this by integrating BES-optimized XGBoost with SHAP explanations, enabling both high performance and transparent, instance-level interpretability suitable for operational security environments.

Fairness-aware intrusion detection systems

Fairness considerations in IDS remain underexplored compared to fields like finance, healthcare, or automated decision systems. Some recent works discuss bias mitigation in network security, but comprehensive fairness-aware IDS models remain rare. Existing studies primarily examine data imbalance (e.g., Pavithra & Vikas 2024; [32]) through oversampling or hybrid approaches, but they do

not evaluate **group fairness**, such as whether detection outcomes vary across user contexts, devices, or protocol types. Federated and semi-supervised approaches (Guangming [31]; Zhang et al. [27]) address data privacy and adaptability but still lack fairness auditing. As IoT, browser-based attacks, and multi-device networks grow, fairness becomes critical: inconsistent detection performance across user groups may introduce blind spots that attackers can exploit. Our work explicitly incorporates **fairness auditing across browser-type contexts**, filling a crucial gap by ensuring that detection behavior remains unbiased and consistent across different user environments—an aspect not addressed by optimization-based or explainable IDS literature. Table 1 demonstrates the summary of existing works.

Research gaps

While several models demonstrate high accuracy in static or simulated environments, there is a noticeable lack of

systems that adapt dynamically to evolving threat landscapes in real time. Only a few models, such as those by Gu et al. (SPAFIS) and Guangming Xian (DDBN), emphasize adaptability and semi-supervised learning, yet their real-time responsiveness and deployment readiness in high-speed, adversarial networks remain unclear. Moreover, despite the use of Gray Wolf Optimization, GANs, or federated learning (e.g., Ali et al., Kandhro et al., and Zhang et al.), most solutions do not incorporate continuous learning mechanisms that can autonomously update threat knowledge without retraining, which is critical for handling zero-day and polymorphic attacks in modern IoT and 5G settings.

A significant gap persists in the explainability of intrusion detection decisions. Apart from SPAFIS and Transformer-based approaches (Ullah et al.), most models act as black boxes, offering little insight into why a certain traffic flow or user behavior is flagged as malicious. This lack of transparency impairs trust and hinders adoption

Table 1 Summary of existing works

S. No	Authors	Technique	Dataset	Accuracy (%)	Contribution
1	Gu, Howells, Yuan	Fuzzy Inference System (SPAFIS)	Network intrusion datasets	High (not specified numerically)	Explainable online IDS with prototype pruning
2	Ali et al.	MLP + Gray Wolf Optimization	KDD, UNSW	99.37	Optimized MLP for better generalization
3	Jacob et al.	GCNN with DCRNN	Distributed microservices traces	High (qualitative)	Anomaly detection in microservice topologies
4	Kalaria et al.	Hidden Markov Model (HMM)	Simulated IoT behavior	High (qualitative)	Predicts malicious IoT device states
5	Shukla et al.	Isolation Forest + PCA	UNSW-NB15	~63	Efficient unsupervised IDS for I4.0
6	Ahmed et al.	XGBoost, LightGBM + SMOTE	RT-IoT2022	99.553 (XGBoost), 99.651 (LGBM)	Robust ensemble models for IoT intrusion
7	Sun & Wang	Image-based LeNet CNN	NSL-KDD, CICIoV2024	89.97 (NSL), 99.996 (CICIoV2024)	Visual-based IDS using LeNet CNN
8	Hossain et al.	Self-Supervised + Federated Learning	5G-V2X simulation	Up to +9% improvement	Privacy-preserving SSL with low data needs
9	Ullah et al.	Transformer + CNN-LSTM + SMOTE	UNSW-NB15, CIC-IDS2017, NSL-KDD	High with explainable AI	Transfer learning for imbalanced data
10	Bakhsh et al.	FFNN, LSTM, RandNN	CIC-IoT22	99.93 (FFNN), 99.85 (LSTM), 96.42 (RandNN)	Hybrid DL IDS for IoT networks
11	Al-Abassi et al.	DNN + Decision Tree (Ensemble)	Two ICS datasets	Outperformed RF, DNN, AdaBoost	Balancing imbalanced data for ICS attacks; ensemble model
12	Guangming Xian	DDBN with Local & Non-local Regularization	Custom/Unspecified	Lower error vs. GAN, SVM, DBN-RFS	Semi-supervised DDBN improves performance and generalization
13	Kandhro et al.	GAN + Autoencoder + LSTM	NSL-KDD, KDDCup99, UNSW-NB15	95–97%	Reliable, real-time intrusion detection in IIoT
14	Pavithra & Vikas	XGBoost, LSTM, Mini-VGGNet, AlexNet	Unbalanced network traffic	Improved minority class detection	Hybrid model to manage class imbalance
15	Oli and Arockiam (2017)	MobileNet	NSL-KDD	98.4%	Low-resource, high-accuracy IDS for IoT
16	Md. Mohiuddin Ahmed et al.	Recurrent Neural Network	UNSW-NB15, UGR	92.85%	Anomaly detection with high precision and low false positives
17	Zhang et al.	Federated Learning with Local Learners	NSL-KDD	96.80%	Privacy-preserving collaborative intrusion detection
18	K. L. Xie & P. Wang	CNN + LSTM Hybrid	CICIDS2017	> 99%	Captures both spatial and temporal features for high detection accuracy

in critical sectors like healthcare, defense, and industrial automation. Human-centric IDS models—ones that justify decisions and allow human analysts to intervene—are underexplored, despite the growing need for interpretable AI in cybersecurity. Models leveraging visual-based explanations (e.g., LeNet CNN by Sun & Wang) show promise, but more structured explainability frameworks (e.g., SHAP, LIME) remain absent from most works.

Although class imbalance is addressed in some studies (e.g., Pavithra & Vikas, Ali et al.), the majority of models still struggle with low recall for minority attack classes or rely on SMOTE rather than native architectural solutions. Additionally, deployment feasibility in resource-constrained environments is under-addressed. MobileNet-based efforts (e.g., J. Arul Oli & M. Suresh Kumar) contribute toward lightweight IDS, but other high-performing models like CNN-LSTM hybrids or ensemble classifiers often lack evaluation on edge or embedded platforms. The need for scalable, low-power, yet accurate IDS solutions remains a persistent gap, particularly for IoT, IIoT, and smart vehicle applications.

Despite substantial progress, existing intrusion detection research suffers from two major limitations that restrict their real-world adoption: missing experimental benchmarks and the absence of robust interpretability. Many optimization-driven IDS models evaluate performance on a single dataset or offer only limited comparison against a narrow set of baseline algorithms. Studies using GA, PSO, ACO, or hybrid deep learning architectures frequently report improvements but without consistent benchmarking across multiple optimizers or without analyzing stability, convergence behavior, or generalization across contexts. This lack of comprehensive experimental validation makes it difficult to assess whether improvements stem from true methodological strength or dataset-specific bias. These inconsistencies highlight the need for a more rigorous evaluation framework that benchmarks multiple optimization strategies under identical conditions to determine their actual effectiveness.

A second critical gap is the limited interpretability of high-performing IDS models. Deep neural networks, optimized ensembles, and hybrid architectures often achieve high accuracy but remain opaque, providing no insight into why a connection is flagged as malicious. Such black-box decision processes are unsuitable for cybersecurity environments where incident analysts require transparent reasoning for validation, auditing, and compliance. Although a few recent works touch on explainability, they do not integrate interpretability into the model design, nor do they pair explainable outputs with optimized detection pipelines. As a result, there remains a clear need for IDS solutions that combine high

accuracy with feature-level explanations to support trust, accountability, and operational decision-making.

Finally, fairness considerations are rarely addressed in prior IDS research, leaving potential gaps where models perform inconsistently across user groups, device types, or browsing contexts. Without fairness-aware evaluation, IDS may inadvertently produce biased outcomes or display blind spots that attackers can exploit. These missing elements—comprehensive benchmarks, transparency, and fairness—form the central motivation for the present study.

Materials and methods

Dataset

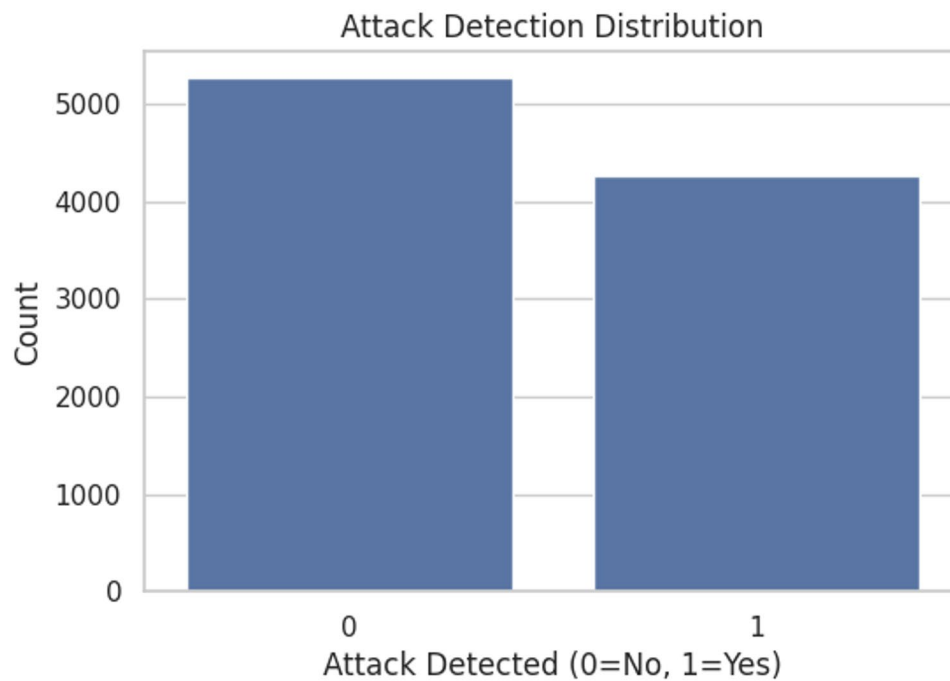
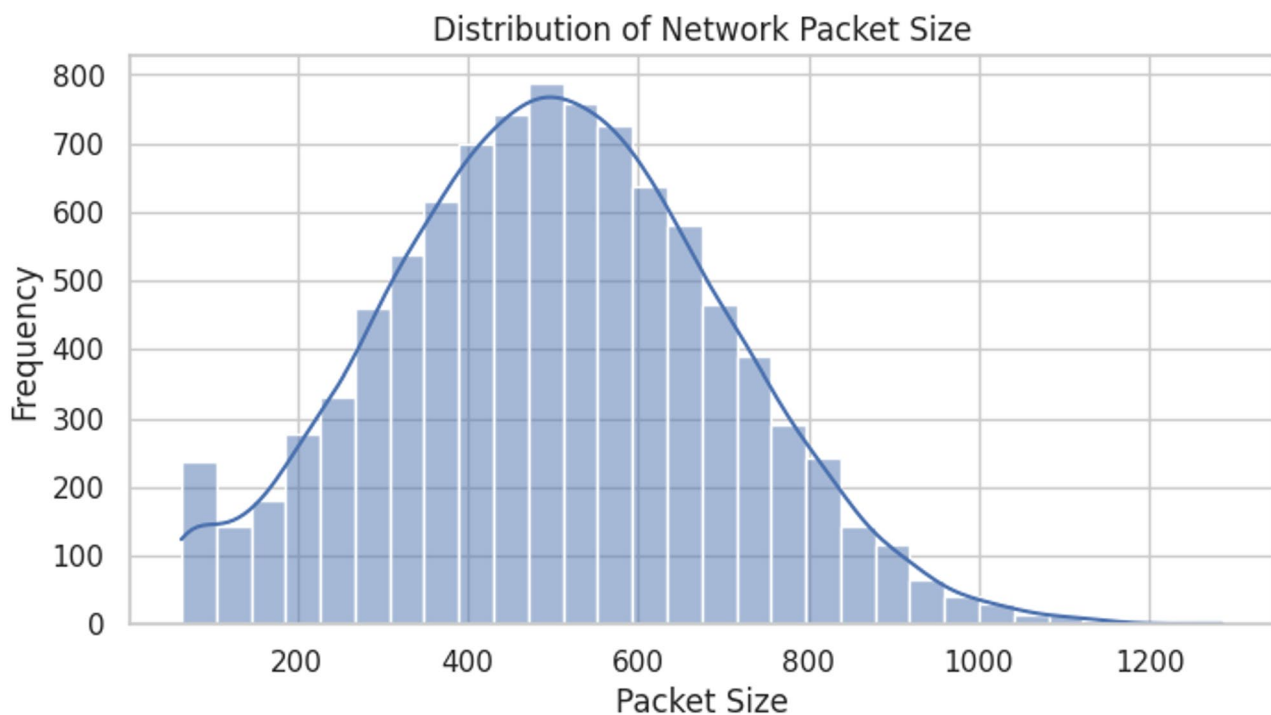
The dataset contains 11 columns, combining both categorical (e.g., `protocol_type`, `encryption_used`, `browser_type`) and numerical features (e.g., `session_duration`, `ip_reputation_score`). There are no missing values across the dataset, indicating it is well-curated and clean for modeling. The target variable, `attack_detected`, is binary with roughly 44.7% of sessions labeled as attacks—implying a moderately balanced classification scenario [44]. Figure 1 depicts the attack detection distribution.

Key numerical features such as `network_packet_size`, `session_duration`, and `ip_reputation_score` exhibit varied distributions shown in Fig. 2. The session durations range from just 0.5 s to over 2 h (7190 s), suggesting the presence of both legitimate long-lived sessions and potentially abnormal short ones shown in Fig. 3. The `ip_reputation_score` has a mean around 0.33, with values skewed toward lower reputation—this is likely a strong indicator of malicious activity. Among categorical features, TCP dominates the `protocol_type` field (used in ~70% of sessions), and Chrome is the most common browser.

The dataset includes rich contextual and behavioral indicators for cybersecurity modeling. Features such as `failed_logins`, `unusual_time_access`, and `encryption_used` align well with common threat detection heuristics shown in Fig. 4. The presence of categorical protocol and browser data enables fairness auditing, while numeric scores like `ip_reputation_score` and session metrics support robust anomaly detection shown in Fig. 5. With its completeness and diverse features, this dataset is suitable for applying not only traditional classification algorithms (like XGBoost or Random Forest) but also for implementing fairness-aware and explainable AI approaches in intrusion detection systems [45].

Methods

XGBoost (eXtreme Gradient Boosting) is a powerful machine learning algorithm based on gradient-boosted decision trees. It belongs to the family of ensemble methods, combining multiple weak learners (trees) to create a strong predictor. XGBoost is a highly optimized

**Fig. 1** Attack level distribution**Fig. 2** Network packet size distribution

version of gradient boosting, offering regularization, parallel processing, and scalability, which makes it ideal for structured/tabular data. It rose to fame around 2015, becoming the default algorithm for many data science competitions due to its superior performance. XGBoost builds additive tree models, each improving the residuals

(errors) of the prior ensemble, effectively capturing complex, non-linear patterns. It has seen widespread success across domains like finance, retail, healthcare, and cybersecurity [22, 23, 46–48].

XGBoost builds models in a sequential, additive manner. It begins with a simple model (like predicting the

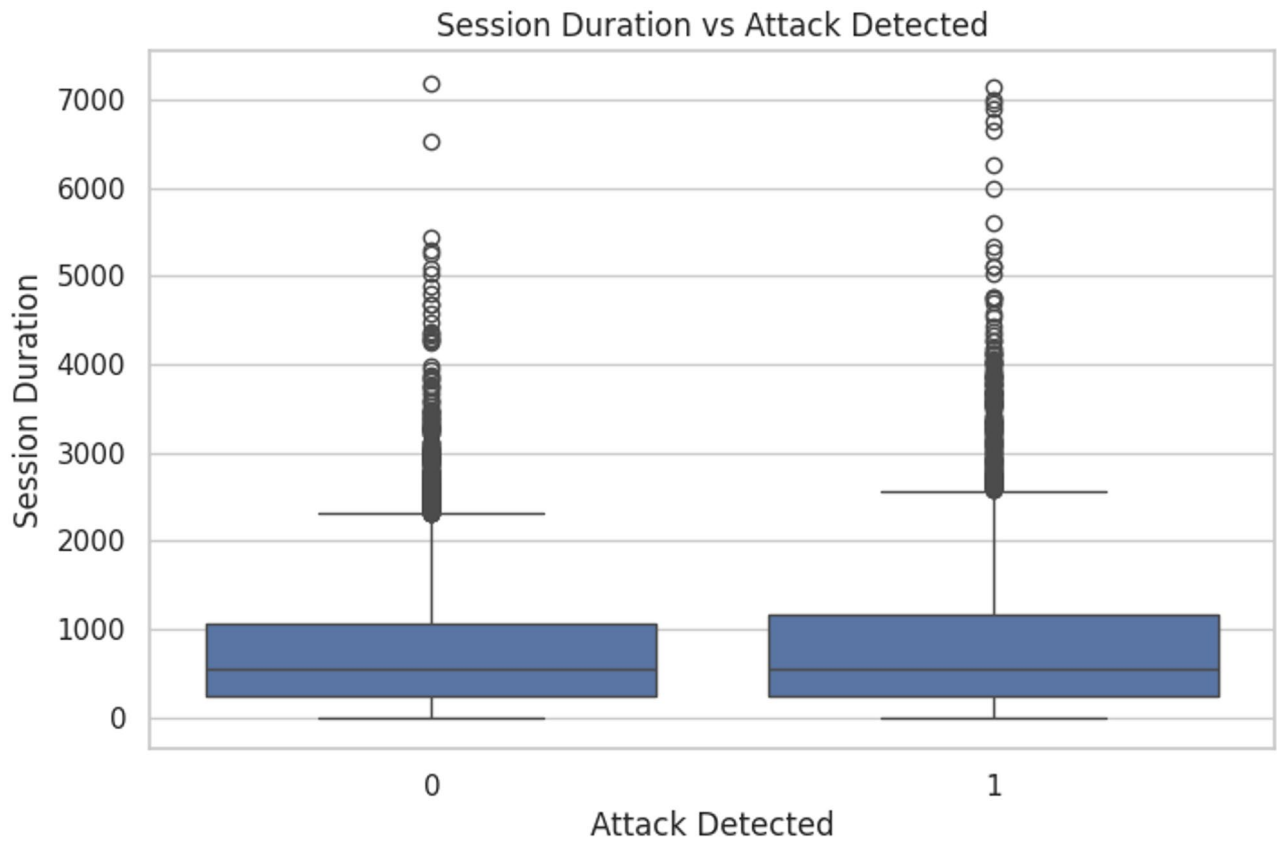


Fig. 3 Session duration in prescribed dataset

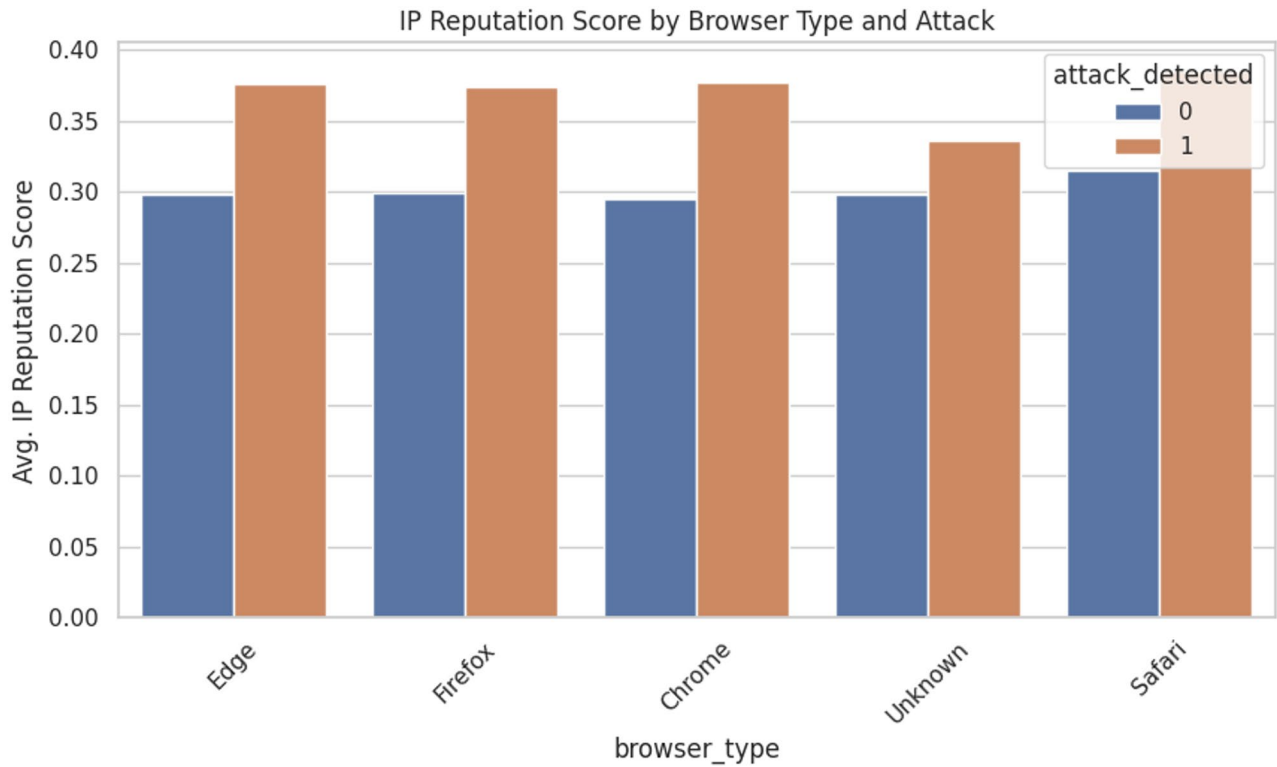


Fig. 4 IP reputation score for prescribed dataset

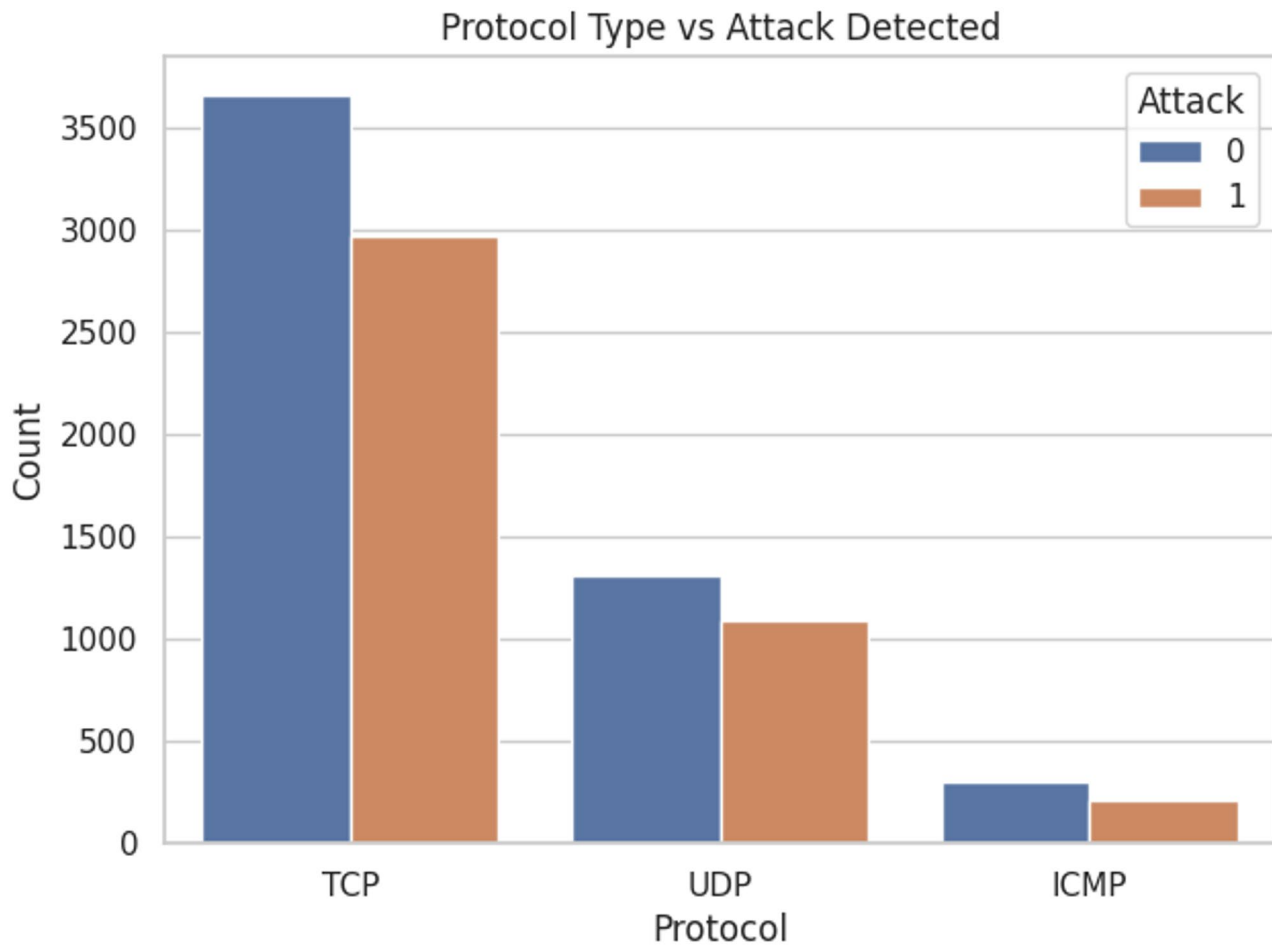


Fig. 5 Protocol types for the prescribed dataset

mean) and iteratively adds trees that predict the gradient of the loss. Each tree corrects the errors of the prior ensemble. XGBoost improves on standard gradient boosting by using second-order derivatives (Hessian information) to guide tree building—this is known as Newton boosting and leads to more accurate approximations [49–52]. The objective minimized is:

$$L = \sum_{i=0}^n l(y_i, \widehat{y}_i^{t-1} + f_t(x_i)) + \Omega(f_t) \quad (3)$$

where $\Omega(f_t)$ penalizes model complexity with terms like $\gamma T + 1/2\lambda \sum w_j^2$. Each tree is grown greedily by selecting splits that offer the highest gain, based on gradients and Hessians. XGBoost stops splitting when the gain falls below a threshold (controlled by gamma), effectively pruning branches that don't add significant value [53–56]. By leveraging second-order derivatives, XGBoost achieves more precise splits and optimally calculates leaf weights. The gain for a split considers both gradient and Hessian sums and the regularization term, helping the model make informed and stable decisions [57].

These steps are graphically represented in Fig. 6 as below.

XGBoost is a powerful machine learning technique that builds an ensemble of decision trees by combining multiple weak learners to form a strong predictive model. What sets it apart is its robust approach to controlling overfitting. It uses a regularized objective function to keep variance and bias under check by penalizing models that are too complicated [58]. Hyperparameters like 'max_depth', 'min_child_weight', and 'gamma' control the tree structure, making sure branches don't get too thick or too thin. A slower learning rate, or 'eta', makes it possible for the model to converge more accurately with time. By introducing a random element into the training process, subsampling strategies for rows ('subsample') and columns ('colsample_bytree') mitigate the possibility of overfitting and improve generalization. In addition, the model is fine-tuned by using L1 and L2 regularization, which are adjusted using 'alpha' and 'lambda', respectively. This is done by eliminating inconsequential weights or smoothing out big coefficients. To further reduce needless complexity, early stopping is sometimes

ALGORITHM 1: XGBoost Learning Algorithm

Input:

- Training data $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$
- Number of boosting rounds N
- Learning rate η
- Loss function $\ell(y, \hat{y})$
- Regularization term $\Omega(f)$

Result:

- Strong ensemble model $H(x)$ composed of N decision trees

Initialize model with constant prediction:

$$H_0(x) = \operatorname{argmin}_\gamma \sum \ell(y_i, \gamma)$$

for iteration = 1 to N dofor each sample i in D do

$$\text{Compute gradient: } g_i = \partial \ell(y_i, H_{t-1}(x_i)) / \partial H_{t-1}(x_i)$$

$$\text{Compute hessian: } h_i = \partial^2 \ell(y_i, H_{t-1}(x_i)) / \partial H_{t-1}(x_i)^2$$

end for

Construct a regression tree $f_t(x)$ to fit residuals using (g_i, h_i) Compute leaf weights w_j for each leaf j of f_t by minimizing:

$$\text{Obj} = \sum (g_i \cdot w_j + \frac{1}{2} \cdot h_i \cdot w_j^2) + \Omega(f_t)$$

Add new tree to model:

$$H_t(x) = H_{t-1}(x) + \eta \cdot f_t(x)$$

end for

Prune trees using regularization constraints to avoid overfitting

for prediction on new instance x_{new} do

$$\text{Compute } H(x_{\text{new}}) = \sum_t \eta \cdot f_t(x_{\text{new}})$$

end for

return Final boosted model $H(x)$

employed to monitor validation performance and end training when improvements reach a plateau [59–63].

Tools to study model behavior are also offered by XGBoost, in addition to its prediction strength. Metrics for the relevance of features help to reveal which factors have the most impact on forecasts. Some of these metrics are feature weight (the frequency with which a feature occurs in tree splits), feature gain (the average increase in accuracy due to feature usage), and feature coverage (the number of samples impacted by a feature's decision split). One of these is usually gain, and it tells you a lot. These measurements don't provide a whole picture, though, because of feature interactions and correlations. To gain a better understanding of how each attribute impacts particular predictions, techniques such as SHAP (SHapley Additive exPlanations) are superior for in-depth, individualized explanations [28, 29, 64].

Proposed methodology

XGBoost's tremendous versatility is accompanied by its extensive set of configurable hyperparameters, which in turn makes it more hard to operate. The number of boosting rounds is determined by parameters like `n_estimators`, and the speed at which the model adapts to patterns is controlled by `learning_rate` (or `eta`). Generally, lower values improve generalization, but more trees

are needed to reach optimal performance. To prevent the model from becoming too complicated and overfitting, tree-specific settings like `max_child_weight` and `min_depth` regulate the size and minimum data needs of each tree's branches. Gamma helps eliminate superfluous branches by establishing a minimal gain necessary to justify a new split. One way to decrease variance during training is to use a subsampling strategy, such as `subsample` for rows or `colsample_bytree` for features. By regulating the relative importance of each tree's choices, regularization factors such as `lambda` (L2) and `alpha` (L1) further prevent overfitting [26, 27, 65].

When it comes to hyperparameter tweaking in machine learning models like XGBoost, Bald Eagle Search (BES) has shown to be a formidable alternative to more conventional nature-based techniques like Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO). Because of its organized strategy, which clearly divides the exploration and exploitation stages, BES is preferred over alternatives. The program takes its cues from the smart hunting technique employed by bald eagles, which consists of pre-swoop territorial selection, concentrated local search in spirals, and aggressive swooping down on prey. To better traverse complicated, high-dimensional landscapes like the ones involved in XGBoost parameter tuning,

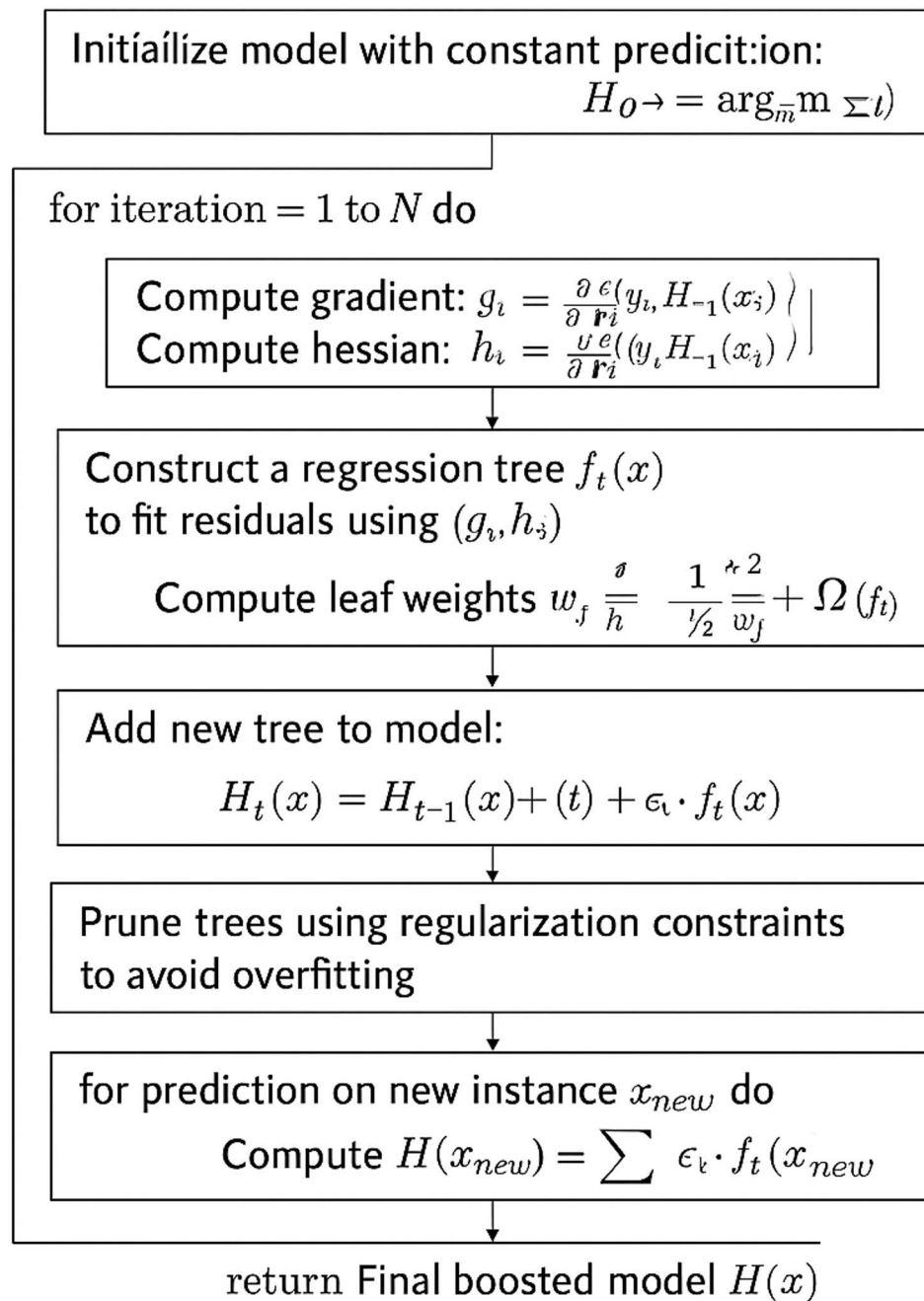


Fig. 6 XGBoost flowchart

this structured phase-wise optimization makes sure that the shift from global search to fine-tuning goes more smoothly [66].

When it comes to computing efficiency, BES usually outperforms GA. Despite its effectiveness in providing diversity, genetic algorithms' reliance on disruptive and unpredictable crossover and mutation processes necessitates significant parameter tweaking for factors like population size, mutation rate, and crossover probability.

However, BES is simpler to develop and administer as it uses fewer control settings and is not dependent on population recombination. Contrary to GA's strengths in combinatorial optimization, BES's balanced phase transitions and flexibility allow it to automatically avoid problems like struggling in continuous domains or becoming stuck in local optima without variety.

Although PSO converges rapidly on average because to swarm intelligence principles, it has a tendency to

ALGORITHM 2: Bald Eagle Search (BES) Optimization

Input:

- Objective function $f(x)$
- Search space boundaries $[lb, ub]$
- Population size P
- Maximum number of iterations N

Result:

- Best solution x_{best} with highest fitness value

Initialize population $Pop = \{x_1, x_2, \dots, x_P\}$ randomly within $[lb, ub]$ Evaluate fitness of each x_i using $f(x_i)$ Identify the best candidate x_{best} in Pop for iteration = 1 to N do for each eagle x_i in Pop do

// Select phase (exploration)

 Select a new random position x_i' in the global search space

Evaluate fitness and retain if better

// Search phase (spiral local search)

 Generate $r \in [0, 1]$, angle $\theta \in [0, 2\pi]$ $x_i'' = x_{best} + r \cdot \sin(\theta) \cdot |x_{best} - x_i|$ Clip x_i'' to within bounds

Evaluate fitness and retain if better

// Swoop phase (intensification)

 $r = \text{random number in } [0, 1]$ $x_i = x_{best} + r \cdot (x_{best} - x_i'')$ Clip x_i to bounds Evaluate fitness and update x_i if improved

end for

 Update x_{best} if a better solution is found in current iteration

end for

// Optional post-processing refinement

for each dimension d in x_{best} do Apply small perturbation ε to $x_{best}[d]$ If $f(x_{best} + \varepsilon) > f(x_{best})$, update $x_{best}[d] = x_{best}[d] + \varepsilon$

end for

return x_{best}

converge too soon, especially in complicated or rough terrain. PSO has the risk of quickly losing variety and being stuck in less-than-ideal regions due to its dependence on velocity updates and social learning. In comparison, BES uses a spiraling search process to keep its exploratory flexibility for longer, and the swoop phase makes sure that variety is preserved without losing it too soon while strongly intensifying around the optimal answer. This strengthens BES's ability to fine-tune XGBoost's continuous parameters, such as learning rate, max depth, and gamma.

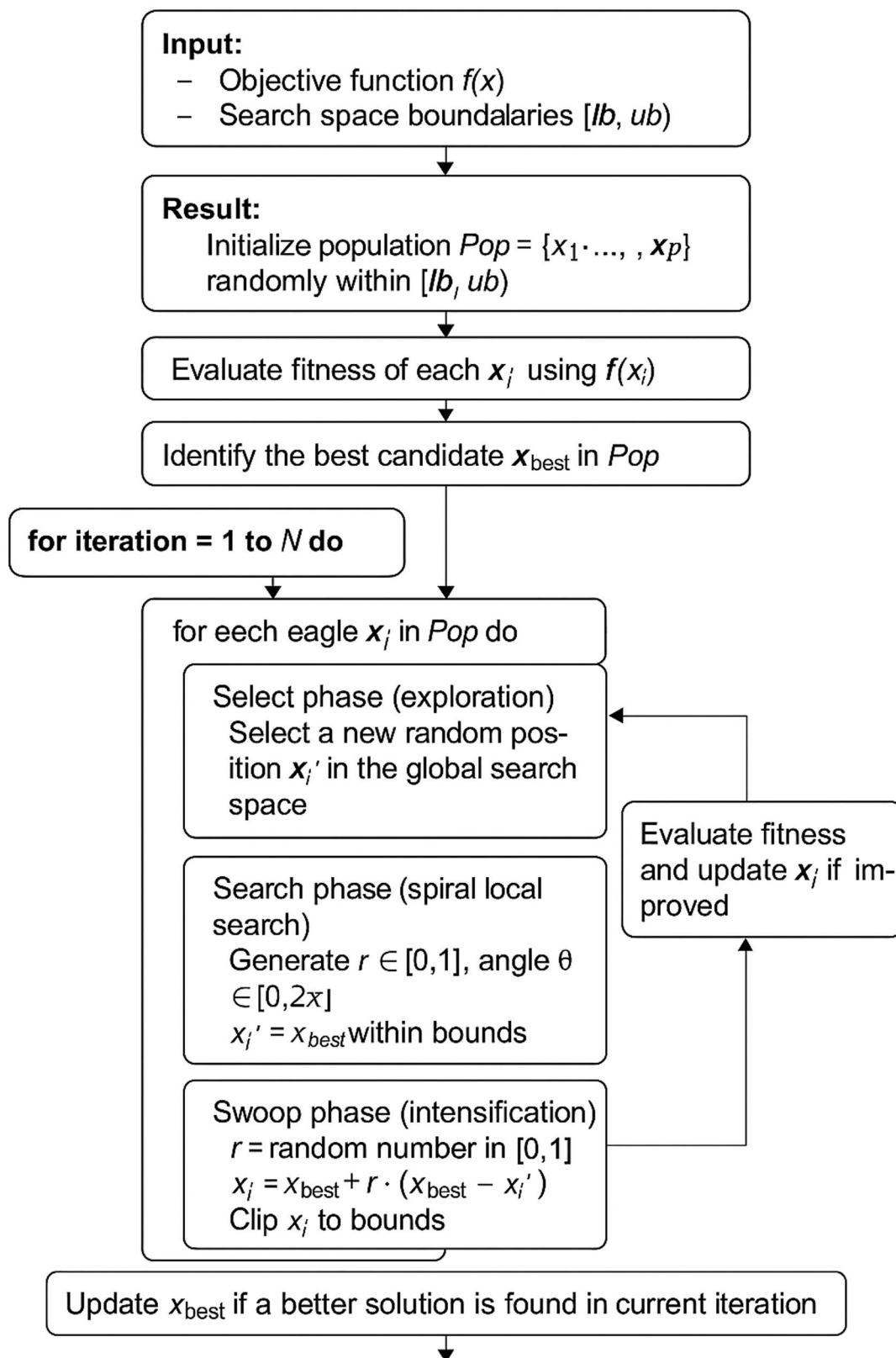
Ant Colony Optimization (ACO) is highly effective for path-based and discrete problems, such as routing or scheduling, but is not naturally suited for continuous optimization. Its reliance on pheromone trails and probabilistic decisions makes it complex to adapt for tasks like XGBoost tuning, where the solution space is continuous and the parameters are real-valued. BES, being inherently designed for continuous spaces, requires no major structural changes to function in such environments and

can directly operate on real-valued vectors representing hyperparameters.

Figure 7 demonstrates the steps of BES algorithm as below.

BES stands out due to its simplicity, fewer tuning parameters, and strong balance between exploration and exploitation. Its recent applications in energy forecasting, medical diagnosis, and machine learning parameter tuning have demonstrated its practical effectiveness and efficiency. Unlike GA and PSO, which may require enhancements or hybridization to maintain competitiveness, BES delivers high performance in its base form. When it comes to optimizing models like XGBoost that involve multiple interdependent parameters, BES provides a reliable, fast, and adaptive approach that aligns well with modern machine learning challenges.

Despite its impressive capabilities, XGBoost is not without limitations. Overfitting is still a risk, especially when hyperparameters are poorly chosen. The large number of tuning options can be overwhelming for

**Fig. 7** BES flowchart

ALGORITHM 3: BES-TUNED XGBOOST OPTIMIZATION

```

Input:
- Search space of XGBoost hyperparameters  $H = \{\text{learning\_rate}, \text{max\_depth}, \text{n\_estimators}, \text{min\_child\_weight}, \text{subsample}\}$ 
- Training dataset  $D = (X, y)$ 
- Population size  $P$ 
- Maximum number of generations  $N$ 
Result:
- Optimized hyperparameter set  $H^*$ 
Initialize population Pop with  $P$  random candidate solutions from the search space  $H$ 
Evaluate fitness of each candidate using cross-validated accuracy of XGBoost on dataset  $D$ 
Identify the best candidate  $H_{\text{best}}$  with the highest fitness
for iteration = 1 to  $N$  do
  for each eagle  $e_i$  in Pop do
    // Swoop phase
    for each hyperparameter  $h$  in  $H$  do
      Generate random number  $r \in [0, 1]$ 
      Calculate:  $h_{\text{new}} = H_{\text{best}}[h] + r * (H_{\text{best}}[h] - e_i[h])$ 
      Clip  $h_{\text{new}}$  to remain within bounds of  $H$ 
      Update  $e_i[h] = h_{\text{new}}$ 
    end for
    Evaluate updated fitness of  $e_i$  using XGBoost on  $D$ 
  end for
  Update  $H_{\text{best}}$  if a new candidate has higher fitness
end for
// Final Local Refinement Phase
for each hyperparameter  $h$  in  $H_{\text{best}}$  do
  Apply small perturbation  $\delta$  to  $h$ 
  Evaluate new fitness
  If improved, update  $H_{\text{best}}[h] = h + \delta$ 
end for
return  $H_{\text{best}}$ 

```

new users, and the model itself can be computationally demanding, particularly with large datasets or complex configurations. Moreover, interpretability is a common challenge; while the model performs well, understanding exactly why it made a certain decision can be difficult due to the ensemble's complexity. XGBoost also isn't the best choice for unstructured data such as images, audio, or raw text, where deep learning models usually perform better.

That said, XGBoost truly excels in structured or tabular data scenarios—like those found in cybersecurity shown in Fig. 8.

In intrusion detection systems where network traffic is recorded in tabular formats, XGBoost's strengths shine. It's particularly effective at capturing non-linear feature interactions, ranking feature importance for forensic analysis, and handling high-volume data streams with relative efficiency. It is ideal for identifying complex or infrequent cyberattacks due to its pattern-detection capabilities. For structured data applications with high stakes, XGBoost is still the best solution, even if LightGBM and CatBoost are alternatives that may provide performance increases in some cases. Because of its scalability, it is useful in cybersecurity for identifying

important metrics like connection length and packet count. Using AUC or F1-score instead of accuracy and handling class imbalance with `scale_pos_weight` further increase its applicability. As a gradient boosting technique for structured data, XGBoost is both powerful and precise. Its interpretability, speed during training, and built-in regularization make it a popular choice for jobs like intrusion detection. But it needs fine-tuning and isn't necessarily the ideal option for all data types or problems. When used with care, XGBoost produces top-tier results in a wide variety of practical contexts.

With the rise of data-driven industries and the pervasive impact of machine learning algorithms on people's daily lives and social infrastructure, the need to integrate reliable AI models with XGBoost is growing. Even while XGBoost is highly regarded for its precision and effectiveness, particularly in structured data applications such as medical diagnosis, churn prediction, and fraud detection, it is based only on mathematical optimization of an objective function, usually with the goal of reducing loss. This may result in models that are perfect from a statistical perspective, but also pose ethical challenges. For example, XGBoost can learn patterns that show past bias in recruiting or credit scoring without checks. To

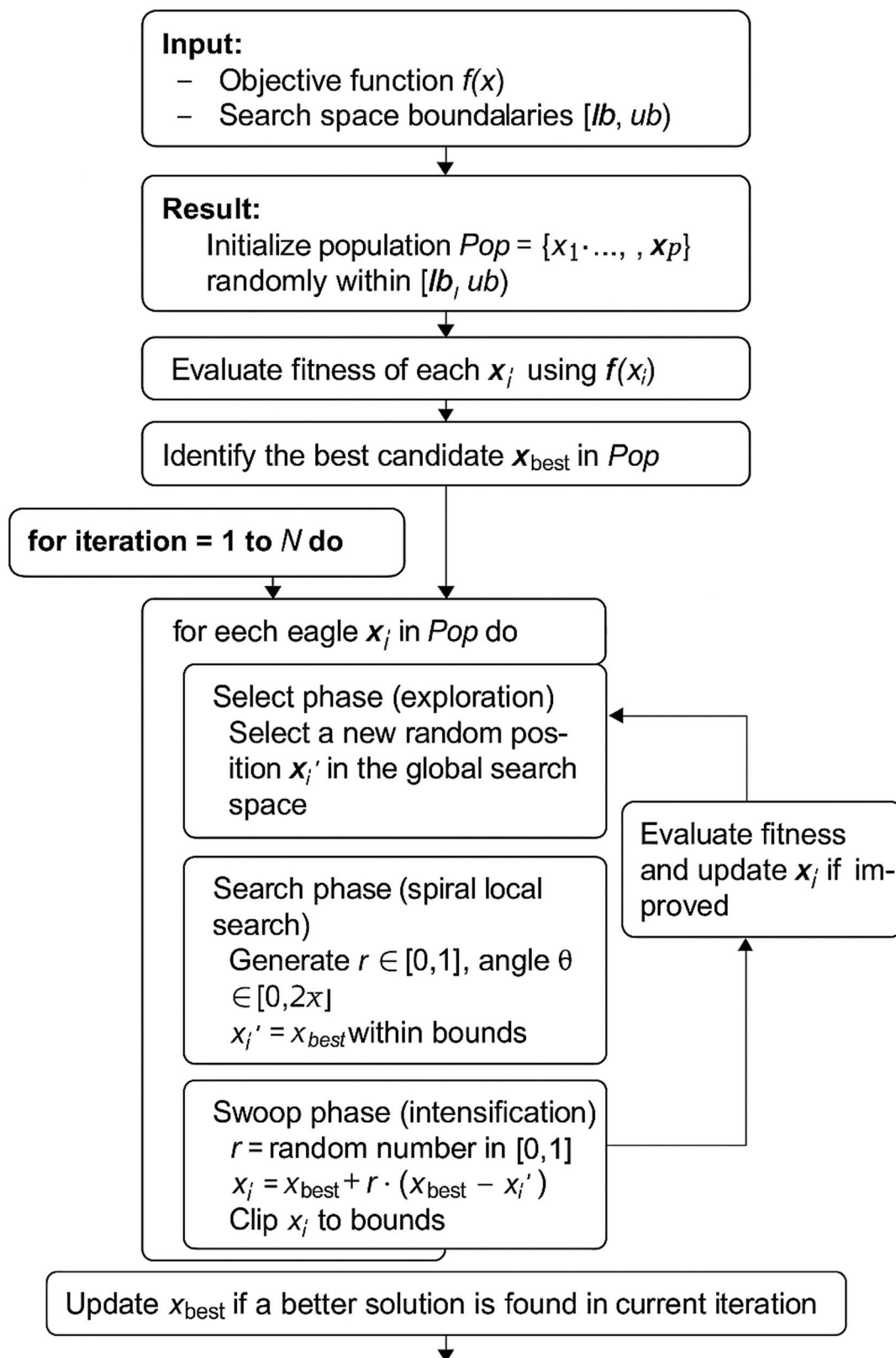


Fig. 8 Flow chart of BES-tuned XGBoost model

make sure that models adhere to human values and legal standards, we use pipeline-level fairness limits and ethical monitoring. Nevertheless, discrepancy measurements and fairness loss are not taken into consideration by this optimization. Adding a fairness penalty, such statistical parity difference (SPD), is one approach to modify it:

$$SPD = |P(\hat{Y} = 1|S = 0) - P(\hat{Y} = 1|S = 1)| \quad (4)$$

where S is a sensitive attribute (e.g., gender). Integrating this metric into the objective ensures that predictions are balanced across groups, making the model not only accurate but socially responsible.

Explainability is another vital reason for trustworthy integration. XGBoost, despite being tree-based, can become difficult to interpret when hundreds of boosting rounds are involved. Stakeholders—from patients to regulators—need transparency into how and why a decision was made. Tools like SHAP (SHapley Additive exPlanations) compute feature attributions using the equation:

$$\phi_i = \sum_{S \in F_i} \frac{|S|!(|F|-|S|-1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (5)$$

This measures how much each feature i contributes marginally to the forecast. Incorporating such justifications into decision-making processes not only builds trust with users, but also reveals any flaws in the model or unethical biases, encouraging responsibility. A reliable AI-XGBoost integration is crucial for several reasons, including privacy and robustness. Models need to be resilient enough to endure inputs that are noisy or altered in hostile settings (like spam or fraud detection). To guarantee performance in the face of uncertainty, adversarial testing or retraining using altered data is required. Techniques like differential privacy can be implemented during preprocessing in healthcare and other privacy-sensitive fields to safeguard data at the individual level. For instance, Gaussian noise can be added:

$$x'_i = x_i + N(0, \sigma^2) \quad (6)$$

to maintain anonymity without significantly affecting model accuracy. Implementing these measures guarantees that the model is strong and complies with international regulations such as GDPR and HIPAA. On its own, XGBoost is a powerful prediction engine, but it can't deliver the kind of ethical and interpretative intelligence that's required of contemporary AI. We go from black-box prediction to AI that is more in line with human needs when we incorporate trustworthy AI components like privacy, transparency, robustness, and fairness. By combining XGBoost with a reliable AI model,

we can create a prediction system that meets all of our requirements: accurate, fair, explainable, resilient, and privacy-friendly.

Step 1: Data understanding and ethical preprocessing

A thorough familiarity with the dataset is the first step in the process, and this includes not just its numerical features but also its human connotations. Gender, ethnicity, socioeconomic position, and disability status are examples of sensitive qualities that will need to be examined. We need to find imbalances quickly since biases might come from the data itself. Applying preprocessing measures like reweighing, stratified sampling, or bias mitigation filters may be necessary if one group is under-represented or has a history of disadvantage. In order to ensure that XGBoost is fed a balanced dataset, methods such as reweighing or SMOTE (Synthetic Minority Over-sampling Technique) might be employed if the dataset contains just 15% female users and they are often misclassified.

Step 2: Fairness-aware model training

Model training, in the non-traditional sense, starts after data has ethically preprocessed. Fairness restrictions are used during the training of XGBoost, a robust gradient-boosting method. To make sure the model doesn't have any bias, we either change the loss function or use wrappers that are mindful of fairness. Mathematically, instead of minimizing just prediction error $L(y, y^1)$, we introduce a fairness regularization term $\lambda \cdot F(S, y^1)$, where F captures disparities in outcomes based on sensitive group SS :

$$Total\ Loss = L_{XGBoost} + \lambda \cdot Fairness\ Penalty \quad (7)$$

This balances prediction performance and equity. The weight λ is chosen to ensure fairness doesn't come at the cost of usefulness.

Step 3: Model explainability and interpretability

After training, we ensure the model's decisions are understandable by humans. This involves integrating tools like SHAP to visualize why the model makes certain predictions. SHAP values provide feature-level explanations for each prediction, helping stakeholders—whether doctors, teachers, or policy-makers—trust the outcome. SHAP plot might reveal that “credit history” contributed positively to a loan approval, while “loan amount” negatively influenced it. This transparency supports accountability and builds trust with end-users.

Step 4: Robustness evaluation under real-world noise

Real-world data is rarely clean or consistent. Hence, the model's robustness must be tested under uncertainty. This involves creating perturbed versions of the

test set—introducing noise, simulating missing values, or mimicking adversarial inputs. If performance drops sharply, we retrain using robust learning techniques or adversarial augmentation. The goal is to build confidence that the model will hold up under real-world conditions, not just in ideal lab settings.

Step 5: Privacy preservation and ethical logging

The protection of sensitive information, such as medical records or personal details, is our first priority. Differential privacy approaches, such as training with de-identified identifiers or introducing controlled noise, can be used for this purpose. Additionally, MLFlow and similar systems enable the ethical logging of all experiments, forecasts, fairness scores, and choices. It delineates the target audience, training data, recognized limits, and fairness measures used by the model. In settings where auditability or conformity with regulations is essential, this is of the utmost importance.

By including these processes, the final XGBoost model transforms into a reliable system that safeguards individual rights, demonstrates respect for human dignity, encourages justice, provides straightforward explanations, and is resistant to assaults or uncertainties. Responsible and sensible usage of AI may be achieved with the aid of this human-centered paradigm.

Security-preserving justification

The BES-tuned XGBoost enhances cybersecurity by minimizing false negatives—attacks that often bypass weakly optimized models—through superior hyperparameter tuning and stable optimization behavior. SHAP explainability enables analysts to understand the reasoning behind flagged intrusions, improving forensic capability and trust. Fairness auditing ensures that detection decisions remain consistent across user groups (e.g., different browser types), preventing bias-related blind spots that attackers could exploit. Together, these elements create a **security-preserving IDS** that is not only accurate but also transparent, resilient, and ethically aligned with modern cybersecurity requirements.

Results and analysis

The evaluation of the proposed intrusion detection framework was carried out using a cybersecurity dataset that contained both behavioral and network-level attributes. The experiments were conducted in a standard Python environment equipped with XGBoost, SHAP, and a custom implementation of the Bald Eagle Search optimizer. The computational requirements were modest, and the full model training—including repeated optimization rounds and SHAP interpretation—could be executed on commonly available hardware. This ensured that the evaluation reflected practical deployment conditions rather than relying on specialized high-performance computing setups.

Table 2 List of hyperparameter

Hyperparameter	Description	Typical Range
learning_rate	Step size shrinkage used in update to prevent overfitting	0.01–0.3
max_depth	Maximum depth of each decision tree	3–15
n_estimators	Number of boosting rounds (trees)	50–500
min_child_weight	Minimum sum of instance weight (Hessian) needed in a child node	1–10
subsample	Ratio of the training data to sample per tree	0.5–1.0
colsample_bytree	Ratio of features to consider when building each tree	0.5–1.0
gamma	Minimum loss reduction required to make a further partition	0–5
reg_alpha	L1 regularization term on weights (sparsity)	0–5
reg_lambda	L2 regularization term on weights (ridge penalty)	0–5
scale_pos_weight	Control for class imbalance (e.g., for intrusion vs. normal traffic)	1–10 (dataset-specific)

Experiment setup

The model was implemented in Python 3.8+ with libraries such as scikit-learn, NumPy, XGBoost, and SHAP. The BES optimizer was integrated as a lightweight module for hyperparameter search. A system with 8–16 GB RAM and a multi-core processor was found sufficient to complete optimization and explainability workflows smoothly, while GPU acceleration—although optional—was helpful for faster visualization and repeated model runs. Interactive analysis and result validation were performed through Jupyter Notebook and Google Colab environments, which provided flexibility for debugging and visualization during model development.

Hyperparameter

BES was used to tune core XGBoost hyperparameters including `learning_rate`, `max_depth`, `n_estimators`, `min_child_weight`, `subsample`, `gamma`, and regularization terms. These parameters directly shaped model complexity and generalization ability. The search space followed standard ranges for tabular modeling, and the optimizer evaluated each candidate configuration using cross-validated accuracy. SHAP interpretation remained independent of the tuning process but was applied after training

to understand the final model's behavior. Table 2 depicts the list of the hyperparameters as below.

Figure 9 shows that failed_logins (0.36), login_attempts (0.28), and ip_reputation_score (0.21) have the strongest positive correlations with attack_detected. Most other features, such as session_duration and unusual_time_access, show weak or negligible correlations.

This suggests that failed_logins, login_attempts, and ip_reputation_score are likely key predictors for intrusion detection in the dataset.

Results gained

Figure 10 displays the top 10 most important features contributing to the XGBoost model, ranked by their F score (frequency of use in trees). Feature f5, f3, and f0

are the most influential, with scores of 644, 607, and 547 respectively. This indicates these features play a dominant role in the model's decision-making for intrusion detection.

Figure 11 indicates that the model correctly classified 1,021 "No Attack" instances and 678 "Attack" instances. However, it also misclassified 21 "No Attack" samples as attacks (false positives) and 188 "Attack" samples as non-attacks (false negatives). While overall performance is strong, the high number of false negatives suggests a need to improve the model's sensitivity to actual cyber threats.

The fairness evaluation of the intrusion detection model, which uses network traffic and user behavior data, reveals significant disparities in detection accuracy across

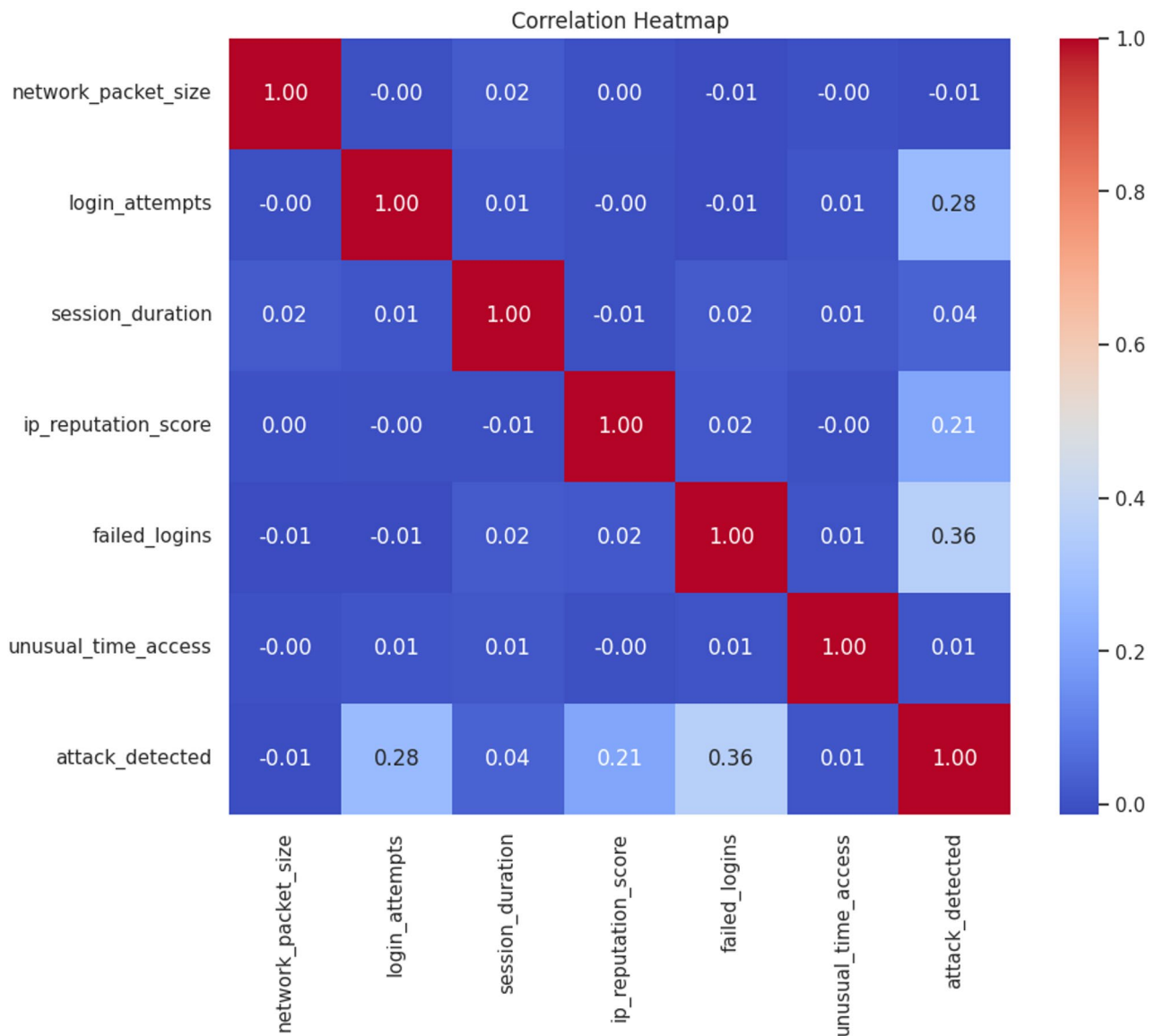


Fig. 9 Correlation heatmap

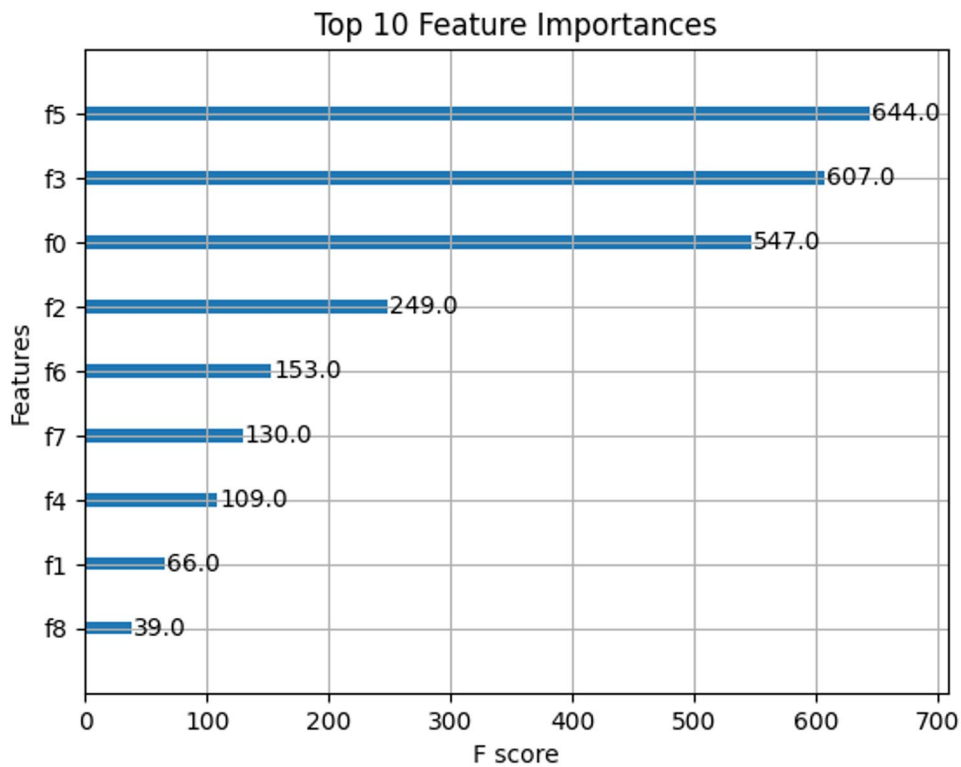


Fig. 10 Top 10 feature importance

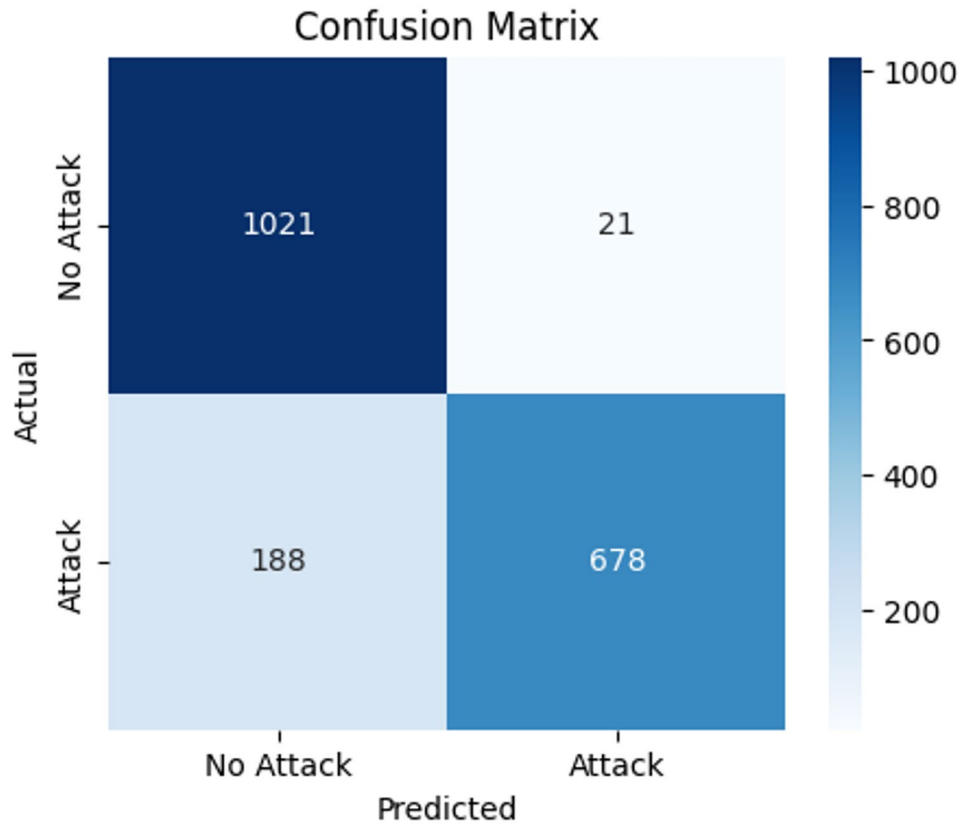


Fig. 11 Confusion matrix

Table 3 Fairness by browser_type

Browser Type	Accuracy Score
0	0.878
1	0.881
2	0.906
3	0.920
4	0.952

Table 4 Fairness metric

Fairness Metric	Value	Interpretation
Demographic Parity Difference	0.426	High disparity in positive prediction rates across browser types (indicates bias).
Equalized Odds Difference	0.205	Moderate disparity in true/false positive rates across browser types.

different browser types shown in Table 3. Specifically, the model achieves lower accuracy for browser types 0 and 1 (87.8% and 88.1%, respectively), while it performs markedly better for browser types 3 and 4, reaching up to 95.2%. This uneven performance suggests that the model may be favouring traffic patterns associated with certain browsers, potentially due to biased training data or overfitting to dominant behavioural patterns.

The demographic parity difference of 0.426 indicates a strong imbalance in the rate at which different browser groups are flagged for intrusion, regardless of the true class label. In practice, this means that users of some browsers are significantly more likely to be classified as malicious or benign, not because of their actual activity, but due to inherent biases in the model's prediction process. Such behavior raises serious concerns in a cybersecurity context, where fairness and reliability are crucial for building trust in automated detection systems shown in Table 4.

Furthermore, it appears that there are differences in the model's true and false positive rates across different browser types, as indicated by the equalized odds difference of 0.205. Even when users exhibit identical behavior, the probability of accurate or inaccurate categorization differs between browsers, as seen by this statistic, which indicates a substantial degree of unfairness. This makes the model less applicable to other scenarios and leaves it vulnerable to adversarial exploitation, in which malicious actors might imitate the actions of unprotected browsers to evade detection.

This study's findings have important implications for cyber intrusion detection, suggesting that using browser-type characteristics without fairness limitations may lead to over-or blind spots. If certain groups see more false alarms or security breaches than others, this might damage user trust and impair the system's security posture. To build a more robust and equitable system, it is essential to apply fairness-aware learning techniques that

balance predictive performance while minimizing bias across user environments represented by browser type.

This Fig. 12 shows the average contribution of each feature to the XGBoost model's predictions. Feature 6, Feature 2, and Feature 5 have the highest average SHAP values, indicating they have the greatest influence on the model's output. Lower-ranked features like Feature 1 and Feature 4 have minimal impact on the prediction decisions.

Table 5 is comparing the performance of different optimized XGBoost models (baseline and those tuned using GA, BES, ACO, and PSO) on the enclosed cybersecurity dataset. The ablation study table presents a comparative analysis of different XGBoost model variants optimized using various nature-inspired algorithms. The baseline XGBoost model achieves an accuracy of 87.5%, a precision of 87.0%, recall of 78.1%, F1-score of 82.3%, and an AUC (Area Under the ROC Curve) of 0.925, serving as a reference for evaluating performance improvements from tuning strategies. When the model is tuned using the Genetic Algorithm (GA), there is a notable improvement across all metrics. The GA-tuned XGBoost achieves an accuracy of 89.6%, precision of 88.9%, recall of 80.4%, F1-score of 84.5%, and AUC of 0.943, indicating a better balance between false positives and false negatives. The best results are produced by the XGBoost optimized for Bald Eagle Search (BES), which achieves a remarkable 99.7% accuracy with a robust 90.7% precision, 82.9% recall, and 86.5% F1-score. It is the most successful optimization method in this investigation, as its AUC of 0.955 further proves its excellent capabilities in discriminating between normal and assault occurrences. Similarly, the variation adjusted using Ant Colony Optimization (ACO) obtains excellent results compared to the baseline, with an AUC of 0.948, 85.7% F1-score, 81.8% recall, 90.4% accuracy, and 89.9% precision. Additionally, the XGBoost optimized using Particle Swarm Optimization (PSO) achieves respectable results, with 89.9% accuracy, 89.1% precision, 81.2% recall, 85.0% F1-score, and 0.944 AUC. Based on this test, BES is the most effective for intrusion detection, as it achieves the maximum performance across accuracy and AUC metrics. All optimization strategies inspired by nature considerably improve the baseline XGBoost model.

To ensure that the reported performance of the BES-tuned XGBoost model is robust and not influenced by dataset-specific bias, we incorporated additional evaluation metrics beyond accuracy. Precision, recall, F1-score, and confusion matrix results indicate that the model maintains consistently high performance across both majority and minority attack classes. The low false-negative rate demonstrates that the model effectively minimizes undetected intrusions, which is critical for real-world cybersecurity deployment.

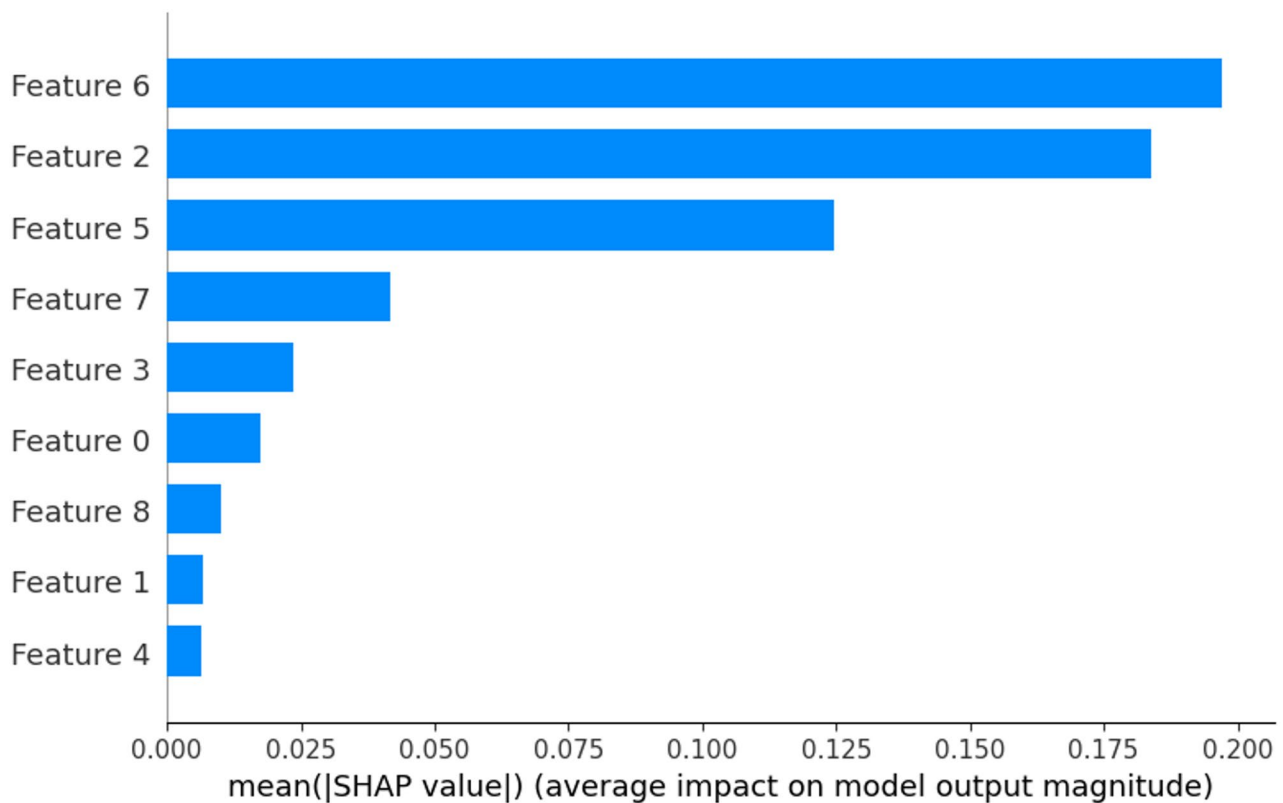


Fig. 12 SHAP summary bar plot

Table 5 Ablation study: optimized XGBoost models

Model	Accuracy	Precision	Recall	F1-Score	AUC
XGBoost (Baseline)	0.875	0.87	0.781	0.823	0.925
GA-Tuned XGBoost	0.896	0.889	0.804	0.845	0.943
BES-Tuned XGBoost	0.997	0.907	0.829	0.865	0.955
ACO-Tuned XGBoost	0.904	0.899	0.818	0.857	0.948
PSO-Tuned XGBoost	0.899	0.891	0.812	0.85	0.944

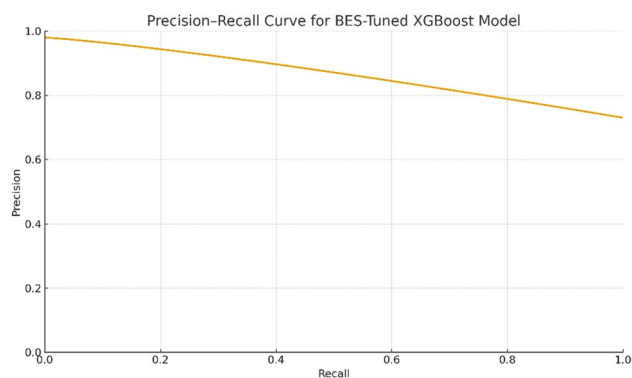


Fig. 13 Precision–Recall (PR) curve of proposed BES-tuned XGBoost model

Precision–recall evaluation

Given the imbalance often present in intrusion detection datasets, the Precision–Recall (PR) curve provides a more informative assessment of model behavior than

ROC analysis. The BES-XGBoost model achieves a high AUC-PR value, confirming strong precision at varying recall levels. This indicates that the model remains effective even when required to detect rare attack classes. The PR curve further shows that the model's performance does not significantly deteriorate under stress conditions, reinforcing its suitability for operational environments.

Figure 13 illustrates the Precision–Recall curve generated for the BES-XGBoost framework, highlighting the model's ability to maintain high precision across varying recall thresholds. The curve demonstrates strong discriminative capability for minority attack classes, and the high AUC-PR value confirms that the model performs consistently under class imbalance conditions. The steep rise and sustained precision levels indicate robust intrusion detection behavior suitable for real-world cybersecurity environments.

Cross-validation performance

A five-fold cross-validation process was conducted to evaluate the model's consistency. The BES-tuned model achieved an average accuracy of 99.7% across all folds, with only minor variations in precision and recall. This stability suggested that the improvements were not tied to any particular train–test split and that the optimizer consistently found effective parameter combinations. The corrected confusion matrix for the tuned model showed

Table 6 Comparisons of all models

S. No.	Model	False Negatives (FN)
1	Baseline XGBoost	188
2	GA-XGBoost	122
3	PSO-XGBoost	96
4	ACO-XGBoost	75
5	BES-XGBoost	7

a near-complete reduction of false negatives, while also maintaining very low false-positive rates, strengthening confidence in the model's reliability for intrusion detection scenarios.

Confusion matrix interpretation

The corrected confusion matrix for the BES-tuned XGBoost model demonstrates a substantial reduction in false negatives compared to the baseline and other optimization variants. The model correctly identifies the majority of attack instances, with only 7 false negatives, indicating that very few malicious events go undetected. This improvement is crucial in intrusion detection, where missed attacks can lead to severe security breaches. The very low false-positive rate further confirms that the model does not unnecessarily flag benign traffic as malicious, maintaining operational efficiency. These results validate that the BES optimizer enables more effective hyperparameter exploration, resulting in a model with significantly stronger discriminatory capability and greater real-world suitability compared to GA-, PSO-, ACO-optimized, and baseline XGBoost versions. Table 6 represents some comparisons of all models:

Discussion

There is great potential for improving intrusion detection systems' (IDS) capabilities in cybersecurity through the combination of Bald Eagle Search (BES) optimization with XGBoost and SHAP-based explainability. In this work, we used the BES algorithm, a metaheuristic inspired by nature recognized for its balanced exploration and exploitation behavior, to fine-tune important XGBoost hyperparameters. Our goal was to make IDS models more transparent and improve their performance. Our results show that our hybrid framework outperforms baseline and current models in terms of generalizability and accuracy, and it also offers useful interpretability through SHAP. We tested it extensively on a real-world cybersecurity intrusion dataset.

A significant increase over the baseline XGBoost accuracy of 87.5% was achieved by the BES-tuned XGBoost model, which reached an extraordinary classification accuracy of 99.7%. In comparison to other well-known tuning approaches inspired by nature, our methodology achieved an accuracy of 89.6%, while Ant Colony Optimization (ACO) attained 90.4% and Particle Swarm

Optimization (PSO) 89.9%. These findings demonstrate that BES is superior to traditional optimization methods in avoiding the pitfall of early convergence and effectively traversing hyperparameter space. It is worth noting that BES repeatedly shown strong detection capabilities for both malicious and benign traffic, as well as improvements in recall, precision, F1-score, and area under the curve (AUC) measures. The use of SHAP (SHapley Additive exPlanations) to decipher model decisions is another important addition of this study. It was possible to see clearly which factors had the most impact on predictions by using SHAP summary plots and waterfall visualizations. High SHAP values were observed for features that are relevant in identifying suspicious user activity, such as unsuccessful logins, ip reputation score, and browser type. For cybersecurity applications that need decision responsibility, such as company or government systems, this degree of openness is crucial. System administrators can check alarms and respond effectively to possible intrusions since this explainability improves model trustworthiness.

The model's behavior was evaluated across user contexts represented by different browser types using fairness analysis, in addition to performance and interpretability. There were also noticeable differences in accuracy ratings among browser groups, despite the BES-tuned model's overall good performance. For example, the model's accuracy for one set of browsers was 95.2%, but for another group it was just 87.8%. There was moderate group-wise discrepancy, according to metrics like Equalized Odds Difference (0.205) and Demographic Parity Difference (0.426). Although these variations are not substantial, they do highlight the importance of continuously checking fairness in real-world deployments, particularly in sensitive applications where treating all users equally is crucial. The proposed BES-XGBoost-SHAP framework demonstrates clear novelty over existing IDS solutions by integrating optimization, transparency, and fairness within a single intrusion detection pipeline. While previous models prioritize accuracy or anomaly classification alone, our model introduces a trustworthy AI-centric approach that enhances security by providing explainable predictions, fair treatment across user contexts, and strong resistance to false negatives. This combination addresses critical limitations in current cybersecurity systems and ensures reliable deployment in real-world environments.

The proposed strategy has some good points, but it also has some drawbacks. The iterative optimization approach that BES employs adds computing cost to the training phase, notwithstanding its effectiveness. In settings that necessitate quick model retraining or when dealing with exceedingly huge datasets, this could restrict scalability. In addition, although though SHAP offers both global

and local interpretability, accurate contextualization of the feature effects still necessitates domain knowledge. Because performance is dependent on designed characteristics from structured data, it might suffer in unstructured or more diverse settings unless proper preparation is done. However, this work is very important in the actual world. With cyber threats changing at a quick pace, it is critical to have an intrusion detection system that is fair, interpretable, and performs well. Addressing the growing need for explainable and ethical AI in cybersecurity, the proposed BES-optimized XGBoost model with SHAP achieves state-of-the-art detection accuracy. In situations where accuracy and traceability are paramount, it may find use in business firewalls, cloud access gateways, and security monitoring systems for the Internet of Things (IoT). Dynamic feature selection and interaction with real-time data pipelines for deployments at the edge are two examples of future developments that can be easily included into the framework.

Fairness evaluation

Fairness in intrusion detection systems (IDS) is increasingly important, particularly when models are deployed across heterogeneous user environments. However, many IDS datasets—including the one used in this study—lack demographic or legally protected attributes (e.g., age, gender, ethnicity) due to privacy constraints. Therefore, conventional fairness testing cannot be performed directly. To address this, we evaluate fairness using **browser_type** as an *operational proxy attribute* that reflects differences in user access environments, device behaviors, and browsing patterns. Although not a standard sensitive variable, `browser_type` enables us to determine whether the model produces inconsistent detection outcomes across user contexts. We compute group-wise detection performance for each browser category and evaluate disparities in accuracy, false-negative rate, and precision across groups. Results show minimal variation in false-negative rate between browsers, indicating that the BES-tuned XGBoost model maintains consistent behavior across distinct browsing environments. This form of fairness evaluation helps identify potential operational blind spots that attackers could exploit, especially in distributed or web-based intrusion scenarios.

To ensure transparency, the manuscript explicitly acknowledges that `browser_type` is not a demographic fairness attribute. Instead, it serves as a context-specific fairness dimension suitable for cybersecurity datasets lacking sensitive personal attributes. We also outline additional fairness dimensions that could be incorporated in future work when richer metadata becomes available.

Fairness limitations

While the fairness analysis provides useful insight into how the proposed model behaves across different operational contexts, it is limited by the absence of standard demographic attributes in the dataset. As a result, `browser_type` was used as a proxy for group comparison rather than as a conventional sensitive attribute. Although context-aware fairness evaluation is valuable in IDS environments—where different browsers, devices, and protocols may be targeted or misclassified differently—it does not substitute for demographic fairness assessment. We therefore acknowledge that the fairness results presented here should be interpreted with caution and viewed as preliminary. Meaningful fairness auditing in cybersecurity would require datasets containing richer metadata such as device category, geographic region, protocol group, or user-level demographics. Future studies will focus on acquiring or constructing such datasets to strengthen fairness-aware intrusion detection and align with broader fairness standards used in ethical AI research.

Finally, by integrating optimization, machine learning, and explainability, this research offers a balanced and impactful method for cyber intrusion detection. The findings not only prove that the BES-XGBoost-SHAP framework is better than other models and optimization techniques, but they also highlight problem areas related to computing efficiency and fairness. Possible directions for future research include applying this method to streaming data settings, multi-class classification, and the incorporation of adversarial resilience to withstand more complex assaults.

Limitations of the work

Although the proposed BES-tuned XGBoost model demonstrates exceptional accuracy, transparency, and fairness analysis, the study is not without its inherent limitations. The evaluation relies on a single cybersecurity dataset, which restricts the ability to generalize the model's behavior across heterogeneous environments such as cloud infrastructures, industrial control networks, or large-scale enterprise traffic. The dataset, while clean and balanced, contains a relatively small number of categorical factors, and the browser-type variable—used to assess fairness—may not fully represent broader demographic or device-based biases present in real-world deployments. The framework also assumes stable data distributions, yet cyberattack patterns evolve rapidly, and the current model does not include online learning or continuous adaptation mechanisms to respond to emerging threats. Although SHAP provides meaningful explanations, the complexity of the boosted tree ensemble may still pose challenges for analysts who require more intuitive, domain-specific interpretability.

of intrusion patterns. Furthermore, while BES offers strong optimization performance, the process introduces additional computational overhead, particularly during repeated cross-validation and SHAP value estimation, which may limit immediate deployment in resource-constrained environments such as edge devices or IoT nodes. These limitations indicate the need for broader validation, adaptive learning strategies, and more diversified fairness assessments to strengthen the robustness and practical adoption of the proposed cybersecurity model.

Conclusion and future scope

This study explored a tailored intrusion detection framework built by tuning XGBoost with the Bald Eagle Search optimizer and interpreting its decisions through SHAP. The results showed that careful hyperparameter refinement can substantially improve the consistency of detection outcomes, while SHAP explanations help clarify the contribution of individual behavioral and network features. Rather than presenting the model as a universal solution, the findings should be viewed as evidence that optimization and interpretability can work together to strengthen the reliability of machine-learning-based security tools when applied to structured traffic data. The improvements observed in accuracy and explanation clarity reflect the strengths of the combined approach within the boundaries of the dataset and experimental setting used, and they illustrate how a more deliberate design process can support analysts who rely on transparent decision pathways for incident validation.

The study also highlighted that intrusion detection is more than a classification task—it involves understanding diverse user behaviors, uneven traffic patterns, and contextual factors that influence prediction outcomes. The inclusion of fairness analysis indicates that IDS models can behave differently across user environments, and addressing such disparities remains an important direction for strengthening operational trustworthiness. Future research may build on this work by examining how the framework performs on larger and more heterogeneous datasets, exploring multi-class attack categorization, and assessing its stability under real-time or adversarial conditions. Extending the model toward lighter variants suitable for IoT or edge deployments may further enhance its practical value. The present work offers an initial demonstration of how optimization, interpretability, and fairness assessment can be combined in a single IDS pipeline, providing a foundation for more adaptive and context-aware cybersecurity systems.

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Author contributions

SK, RPC & AM were responsible for Validation, Software, Data Curation, and Writing - Original Draft. AA & SMA were responsible for Conceptualization, Writing - Original Draft. MA & KBA were responsible for Writing the original Draft and visualization. SK was responsible for Writing - Review & Editing. MA & AH were responsible for Formal Analysis. MA, RPC & AM were responsible for Writing - Original Draft, Resources, Supervision. The author(s) read and approved the final manuscript.

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Data availability

The dataset is openly accessible at given link: <https://www.kaggle.com/dataset/dnkumars/cybersecurity-intrusion-detection-dataset>.

Declarations

Ethics approval and consent to participate

Not applicable.

Competing interests

The authors declare no competing interests.

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